

MACHINE LEARNING FOR SIGNAL PROCESSING

19-3-2025

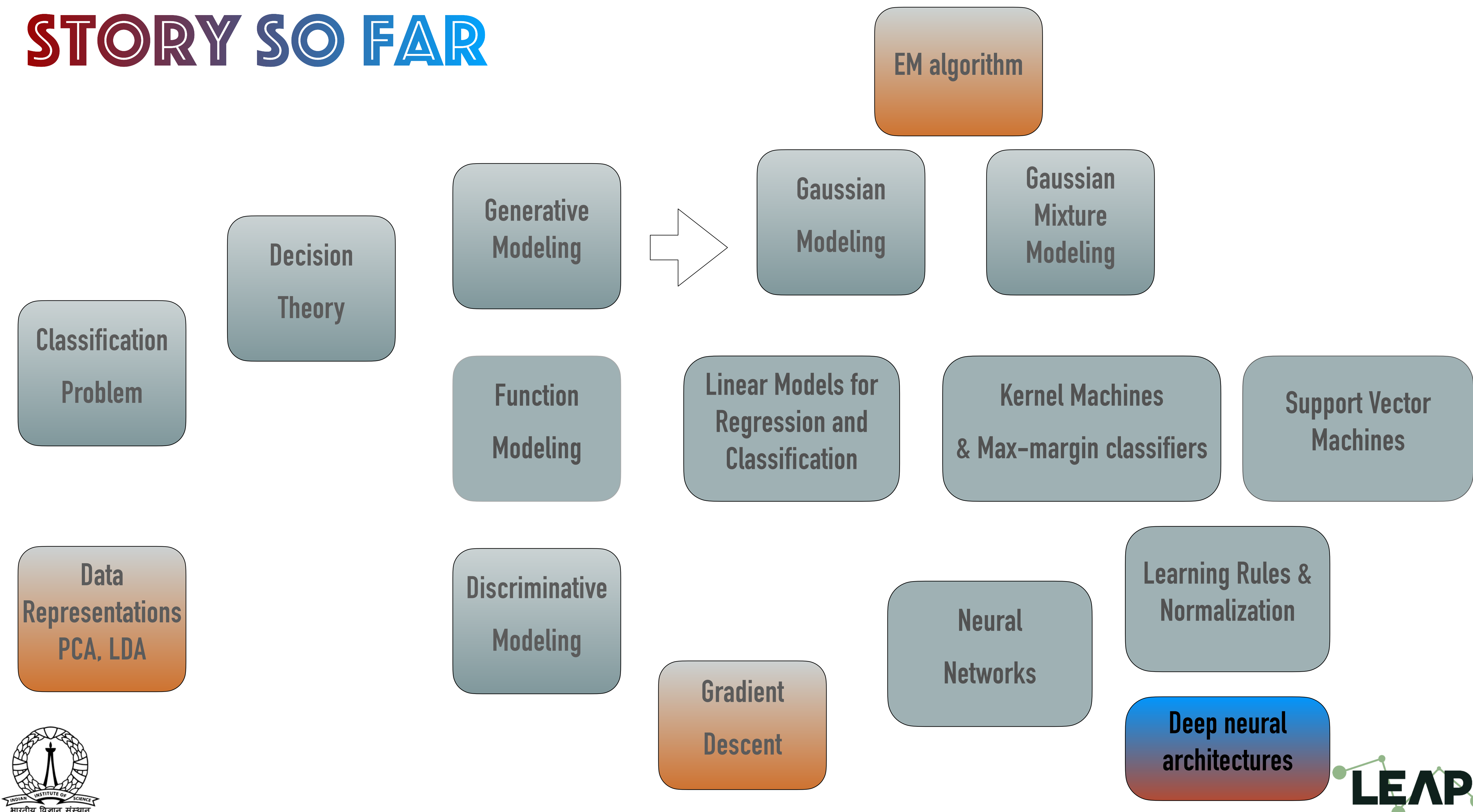


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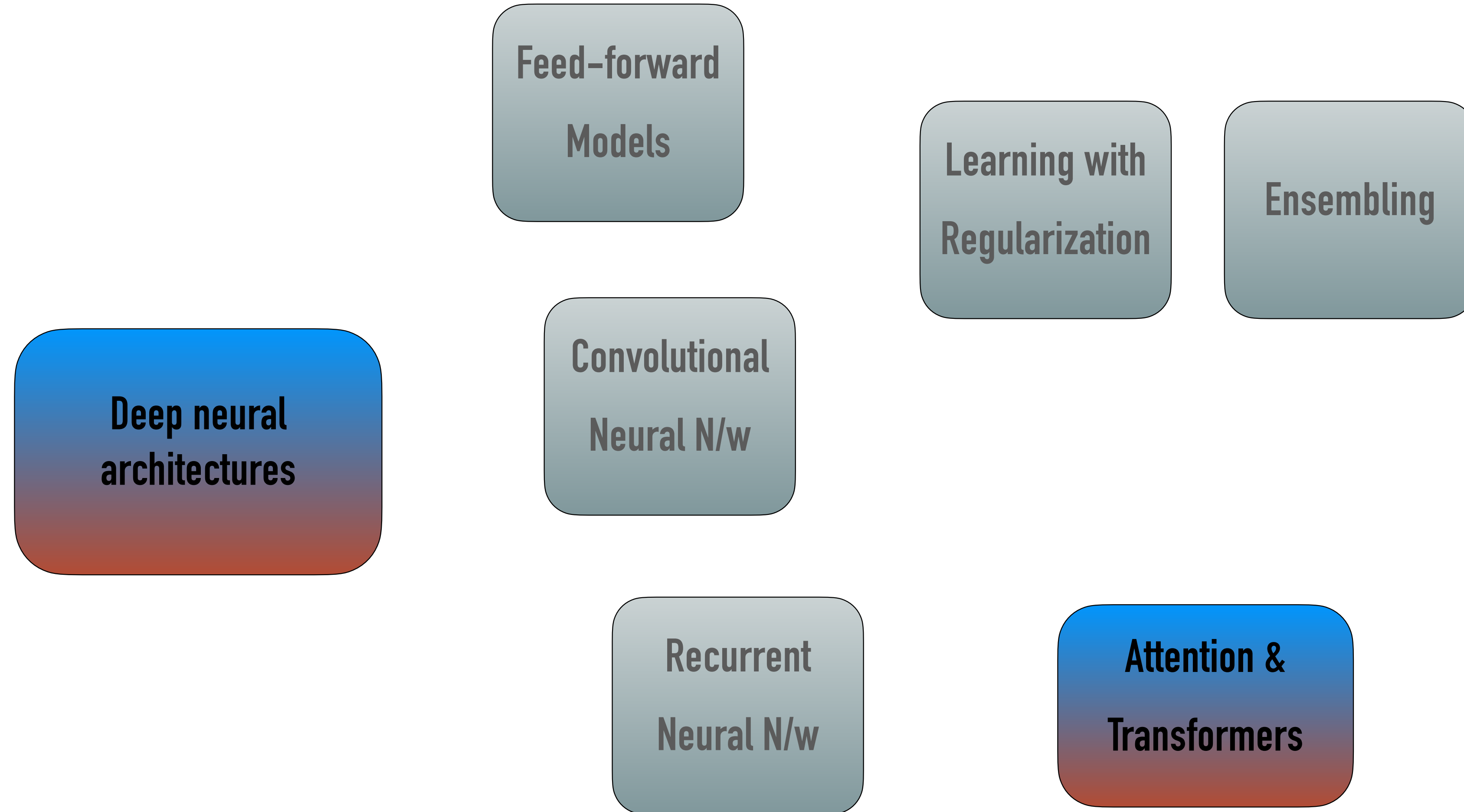
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Viveka Salinamakki, Varada R.
LEAP lab, Electrical Engineering, Indian Institute of Science
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<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>

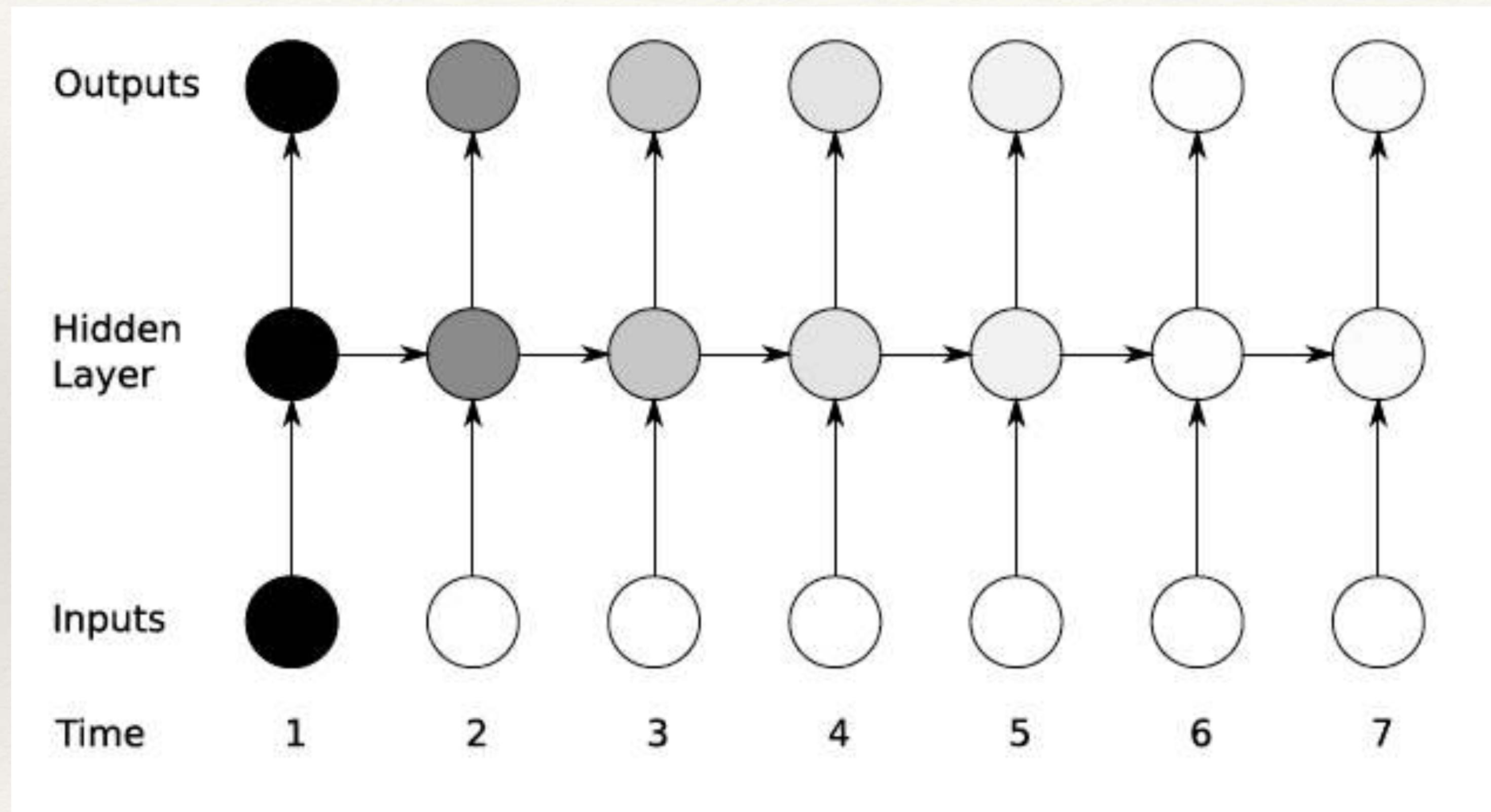
STORY SO FAR



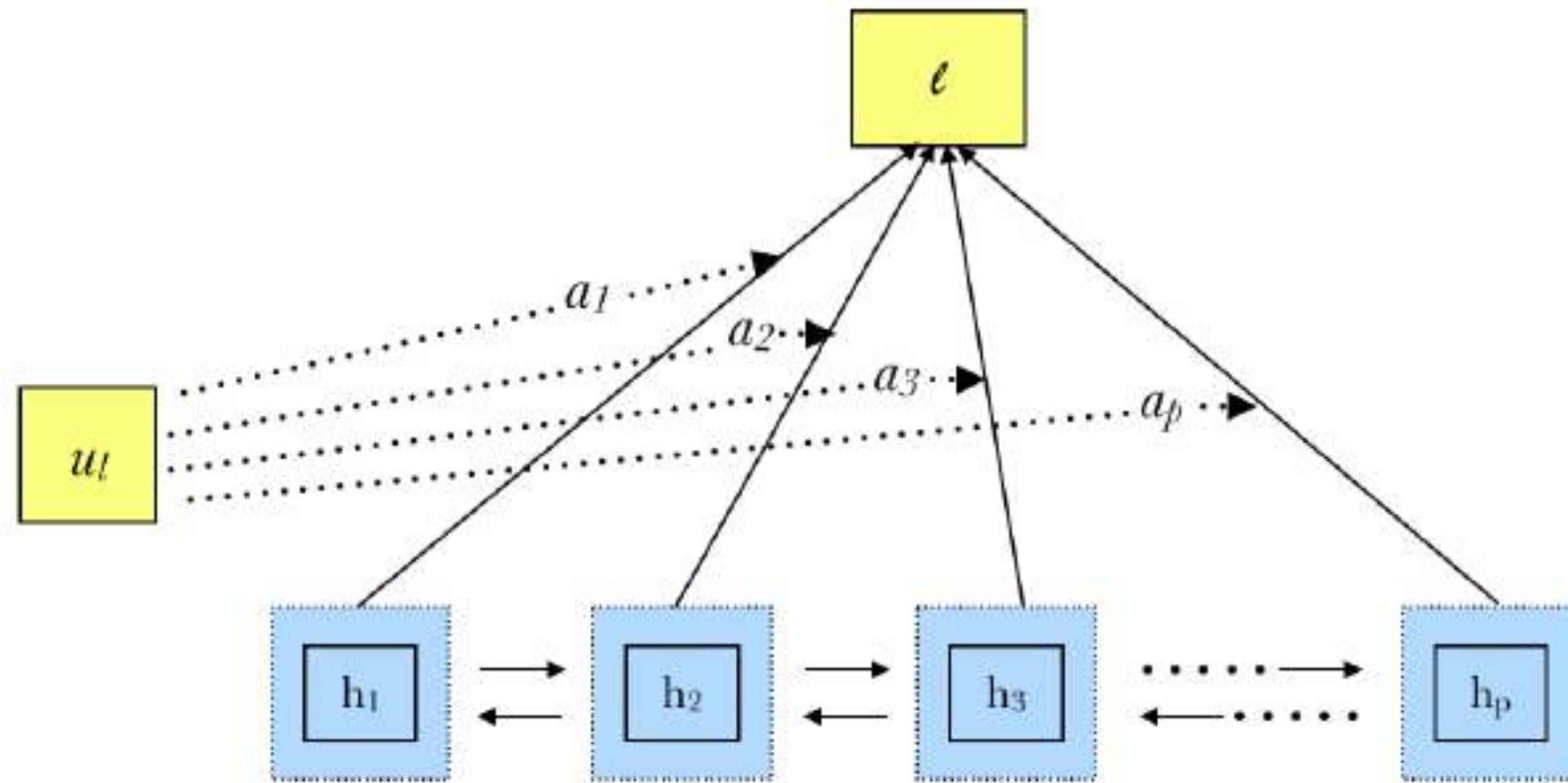
STORY SO FAR



Long-term Dependency Issues



Attention in LSTM Networks



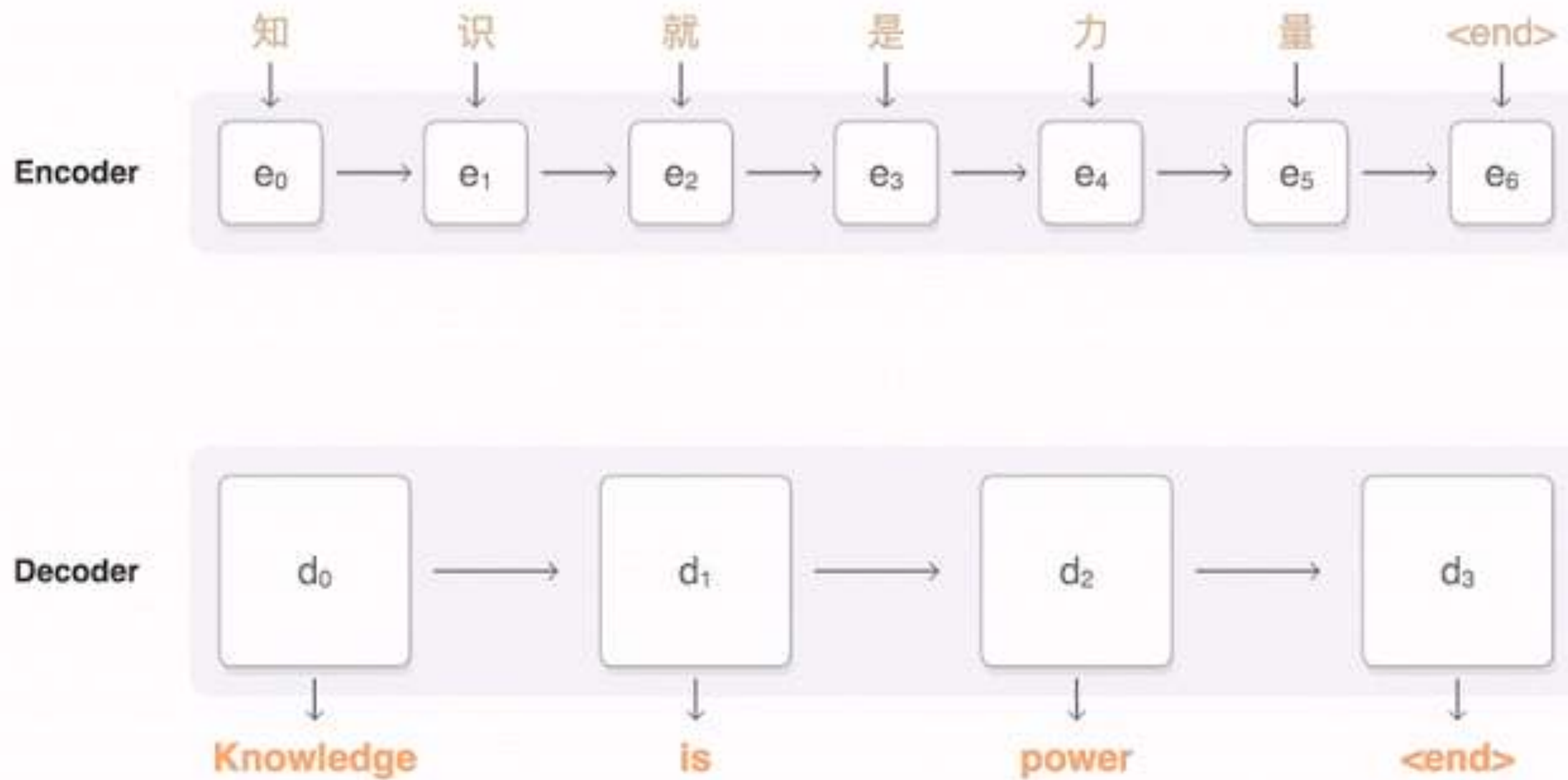
$$\mathbf{u}_t = \tanh(\mathbf{W}_l \mathbf{h}_t + \mathbf{b}_l)$$

$$a_t = \frac{\exp(\mathbf{u}_t^T \mathbf{u}_l)}{\sum_t \exp(\mathbf{u}_t^T \mathbf{u}_l)}$$

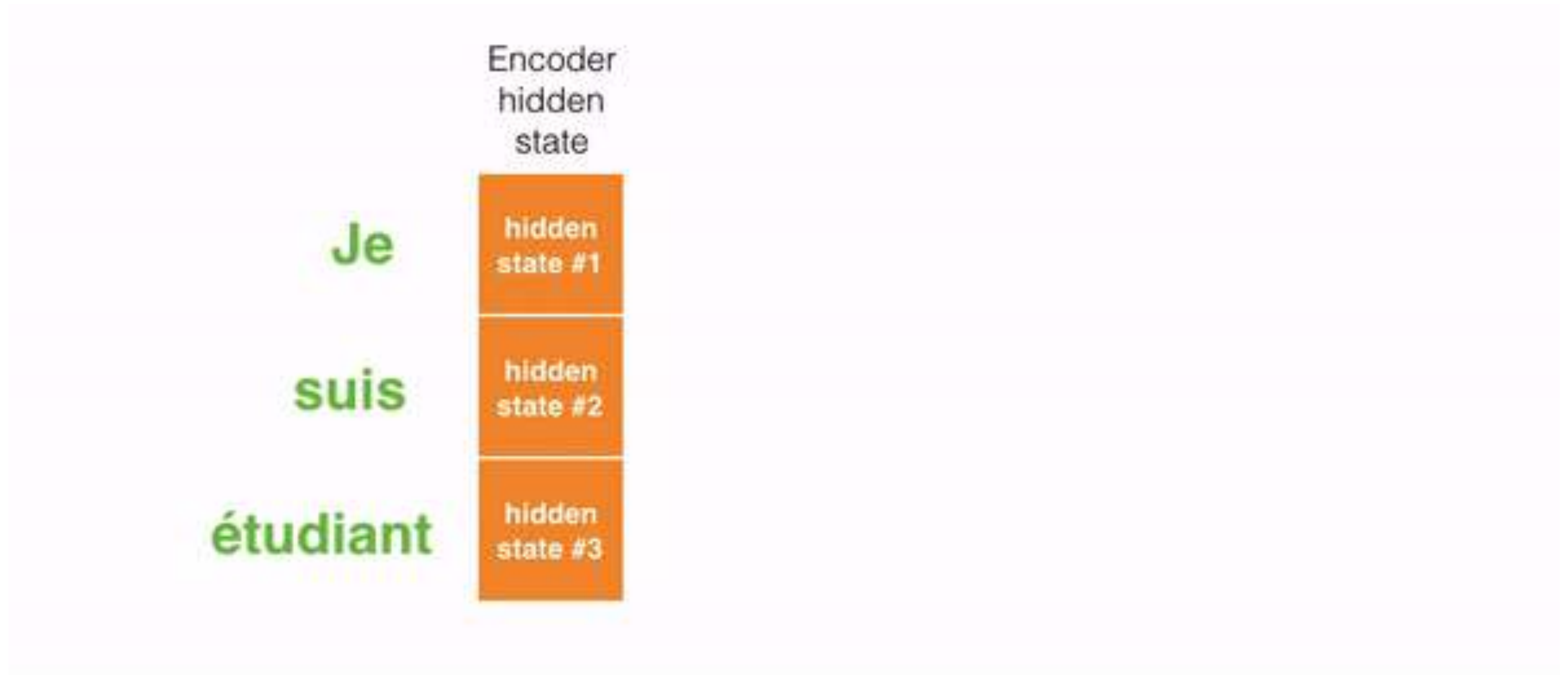
$$\mathbf{l} = \sum_t a_t \mathbf{h}_t$$

- ❖ Attention allows a mechanism to add relevance
- ❖ Certain regions of the audio have more importance than the rest for the task at hand.

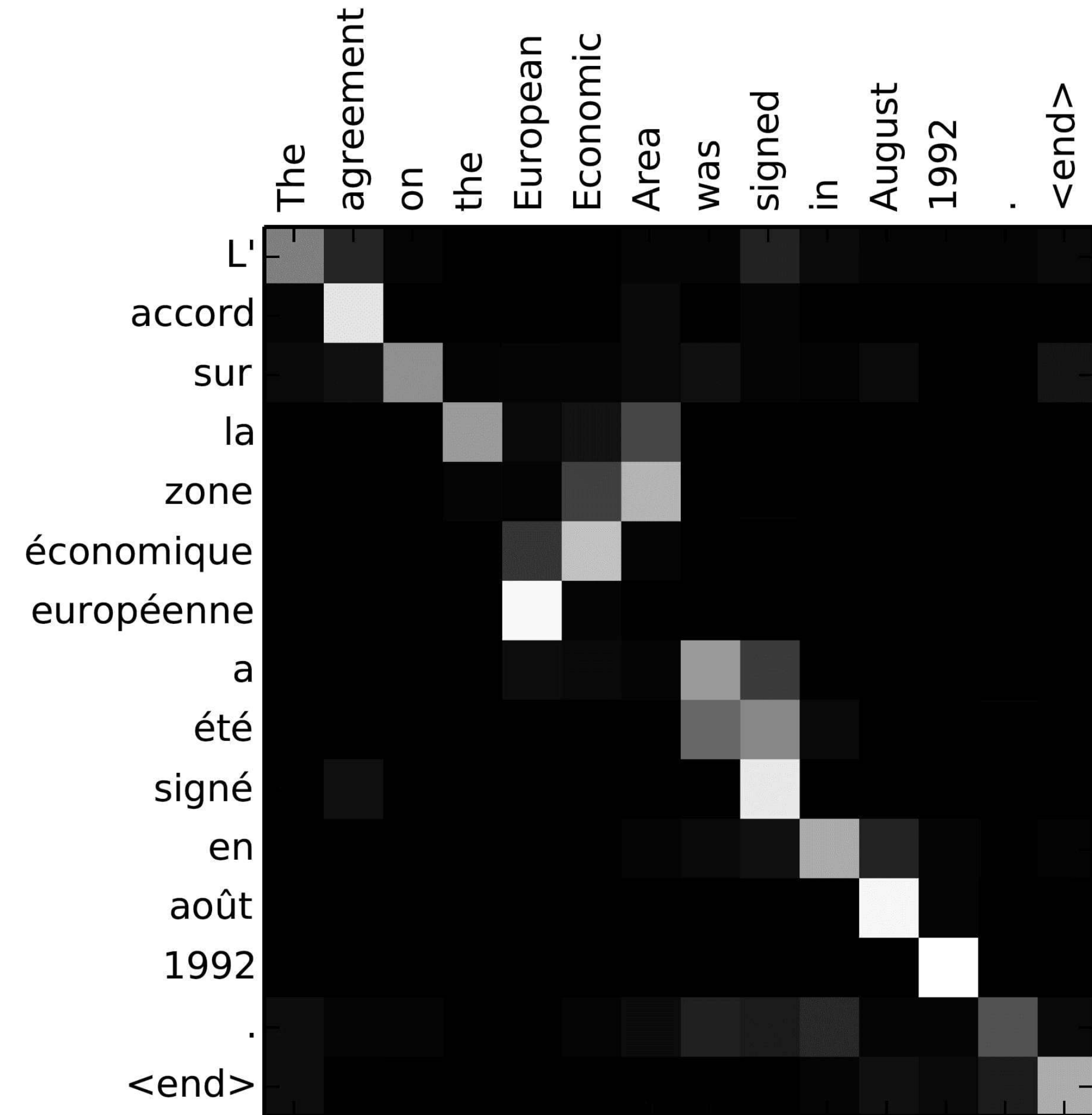
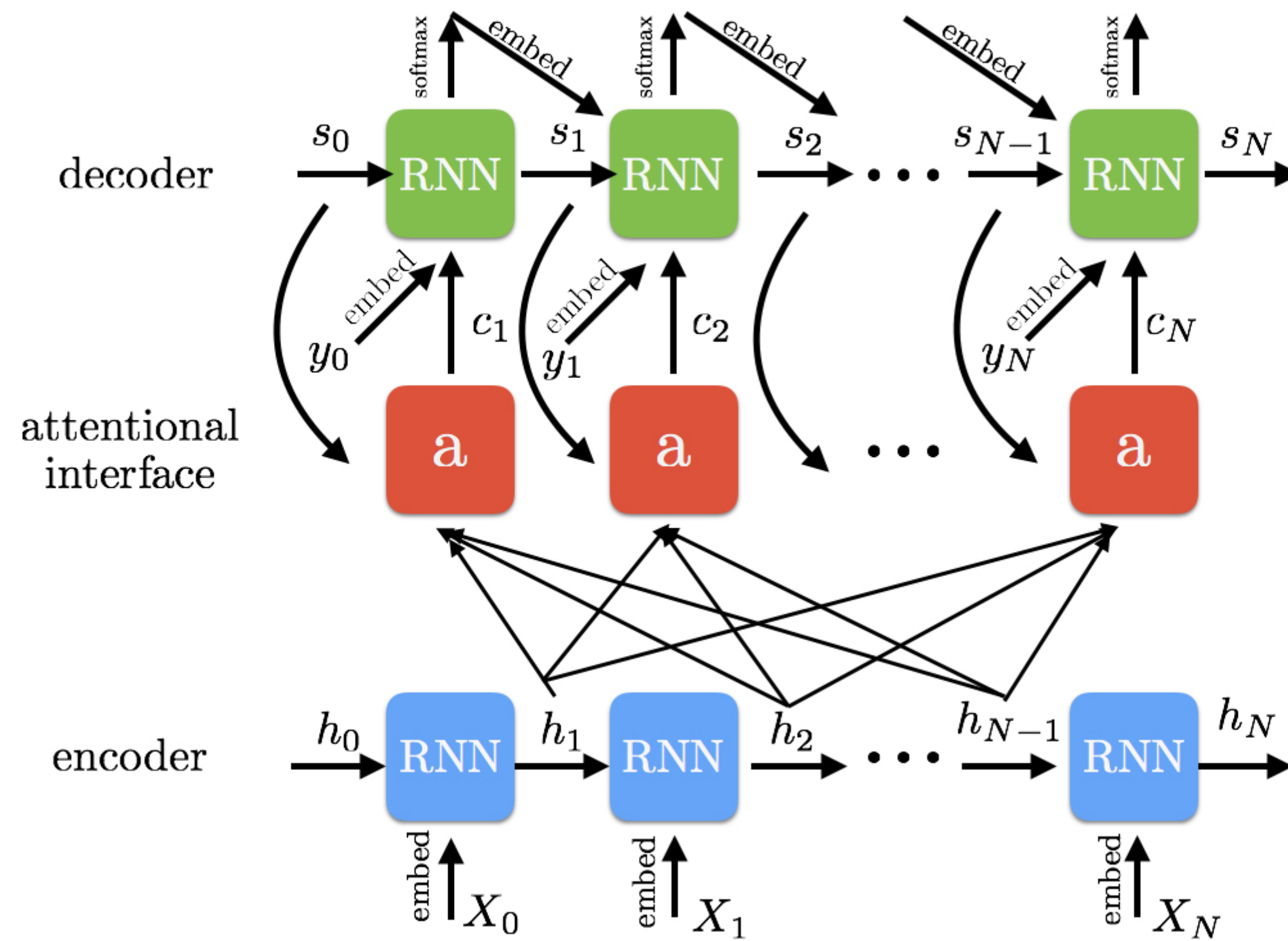
Encoder - Decoder Networks with Attention



Encoder-Decoder Attention

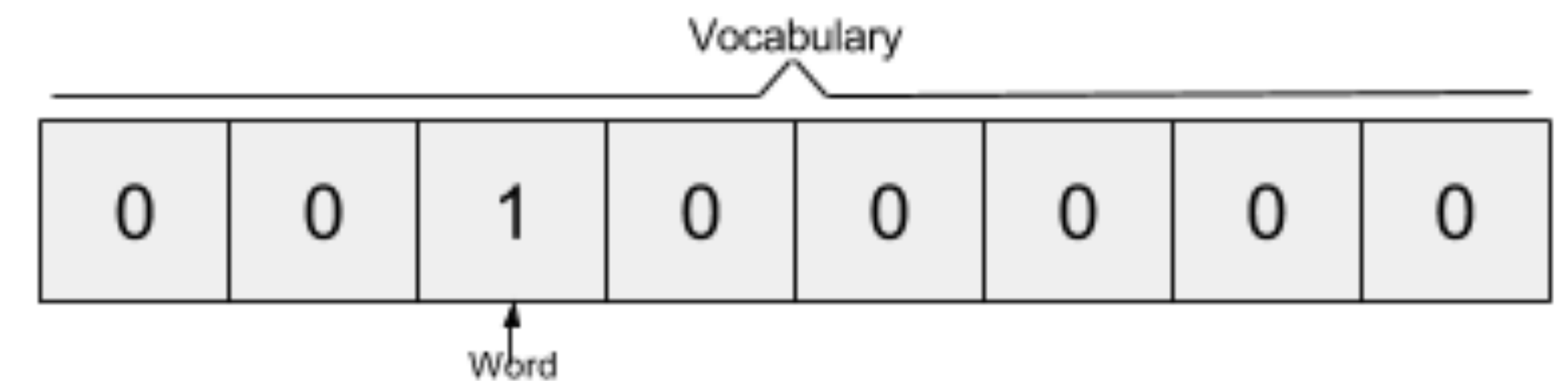


Self-Attention Models



Natural language processing - representations of text

- ❖ Converting text into fixed representations in a vector space.



- ❖ Using one-hot vectors is really high dimensional.
- ❖ Need a concise word representations that embeds semantics.

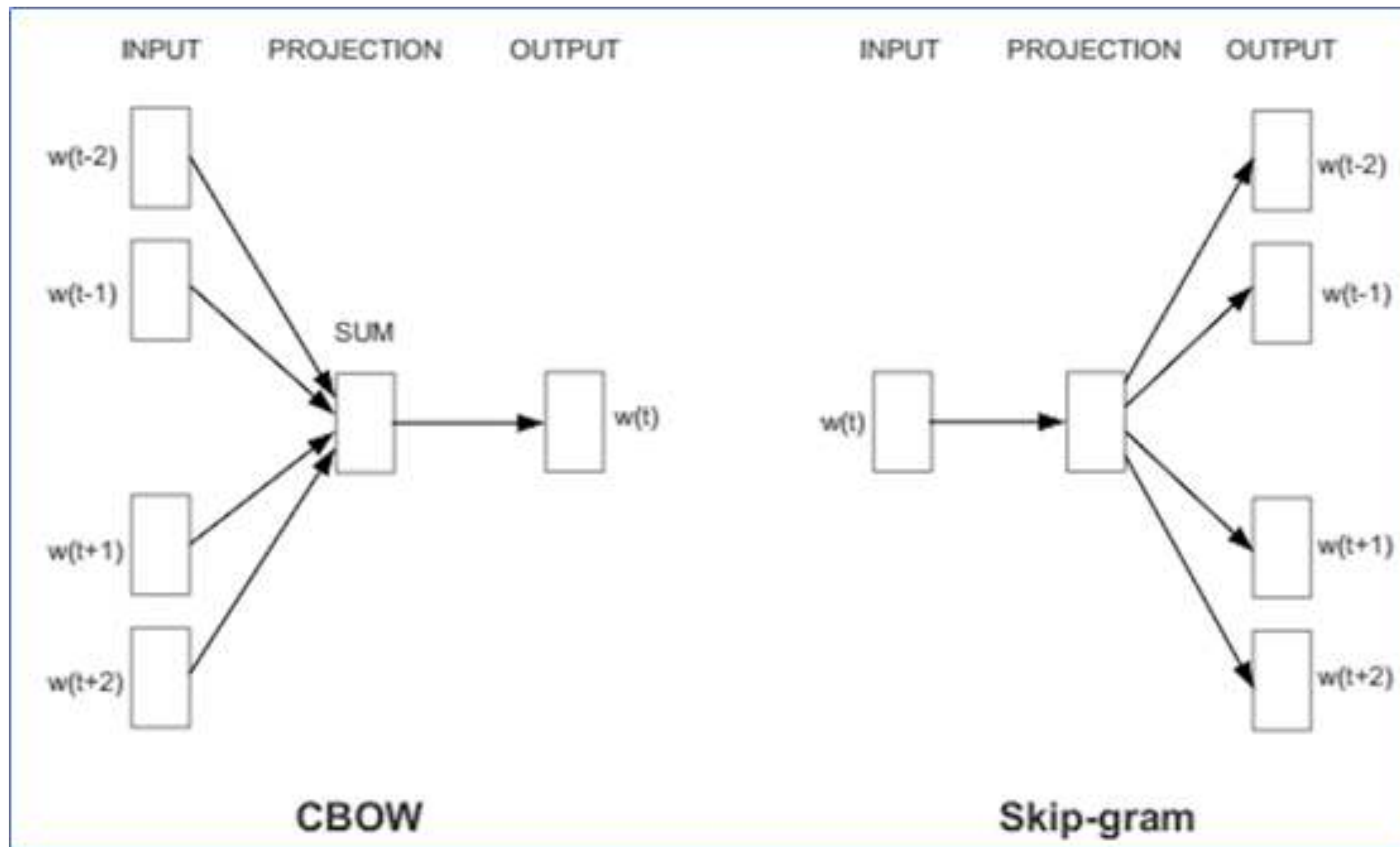
word2vec models as text representations

Source Text	Training Samples
The quick brown fox jumps over the lazy dog. ➡	(the, quick) (the, brown)
The quick brown fox jumps over the lazy dog. ➡	(quick, the) (quick, brown) (quick, fox)
The quick brown fox jumps over the lazy dog. ➡	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
The quick brown fox jumps over the lazy dog. ➡	(fox, quick) (fox, brown) (fox, jumps) (fox, over)

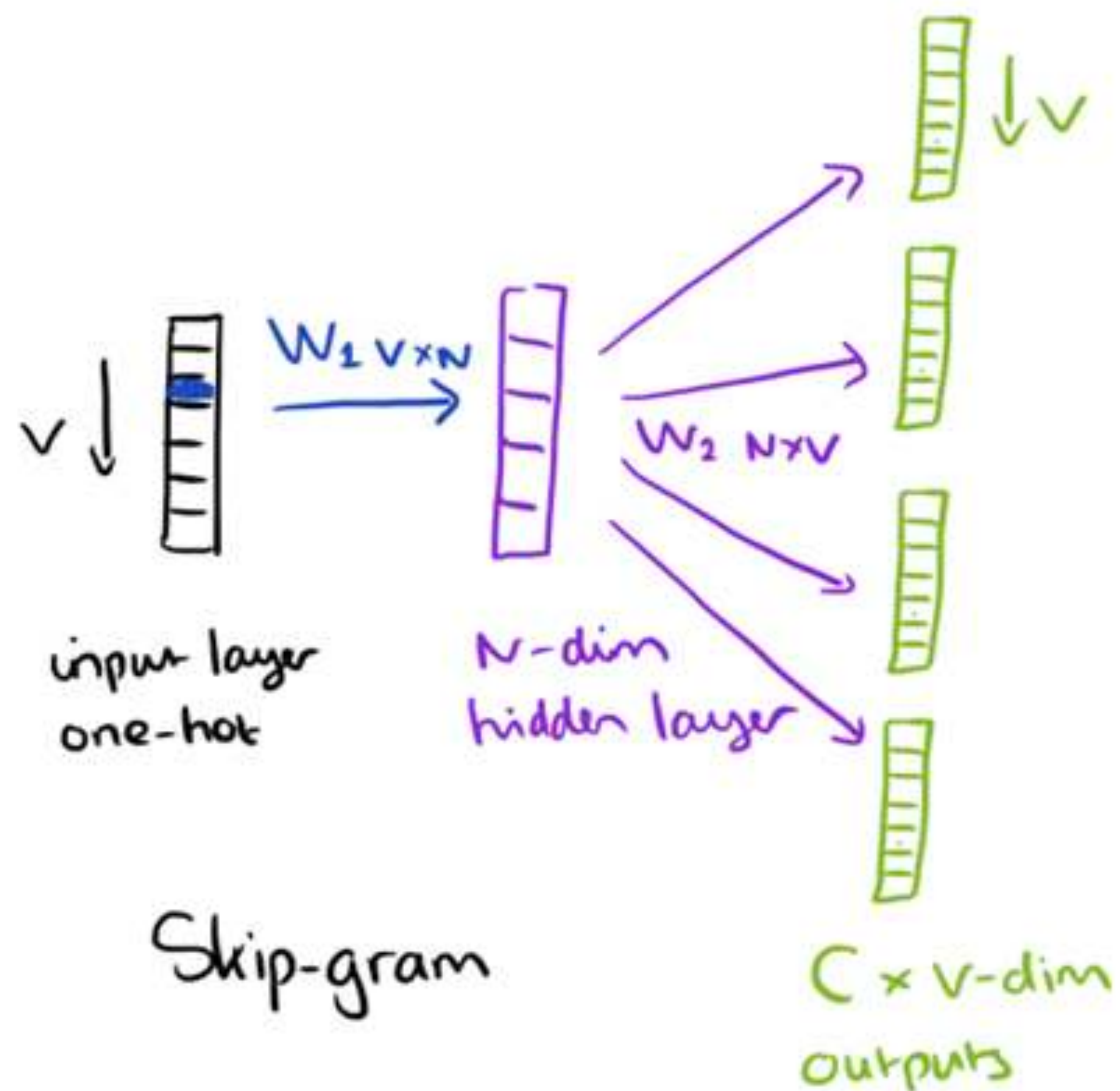
word2vec representations



word2vec models as text representations



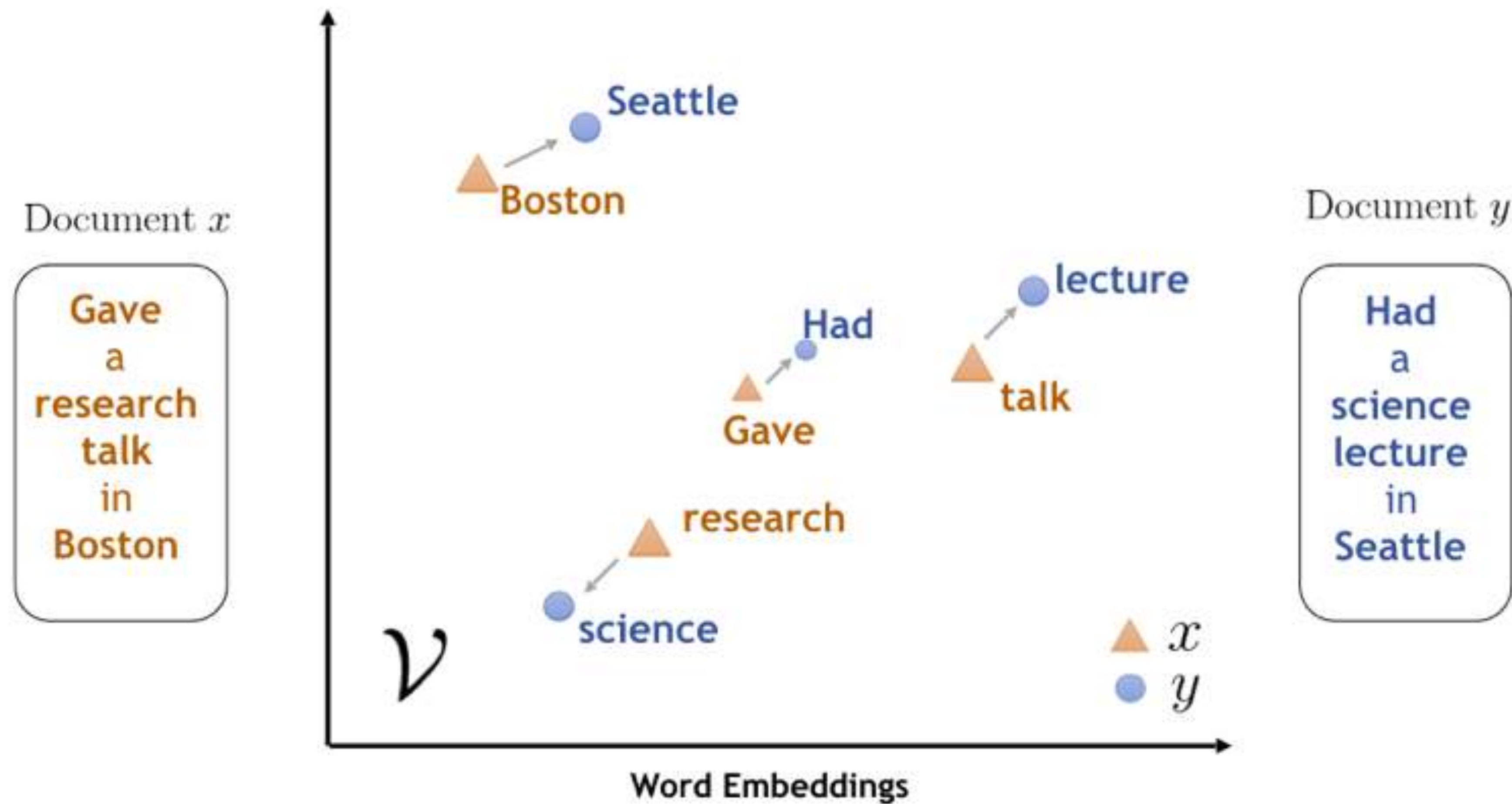
word2vec representations



$$\begin{array}{c} \text{input} \\ 1 \times V \end{array} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{array}{c} W_1 \\ V \times N \end{array} \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{bmatrix} = \begin{array}{c} \text{hidden} \\ 1 \times N \end{array} \begin{bmatrix} e & f & g & h \end{bmatrix}$$

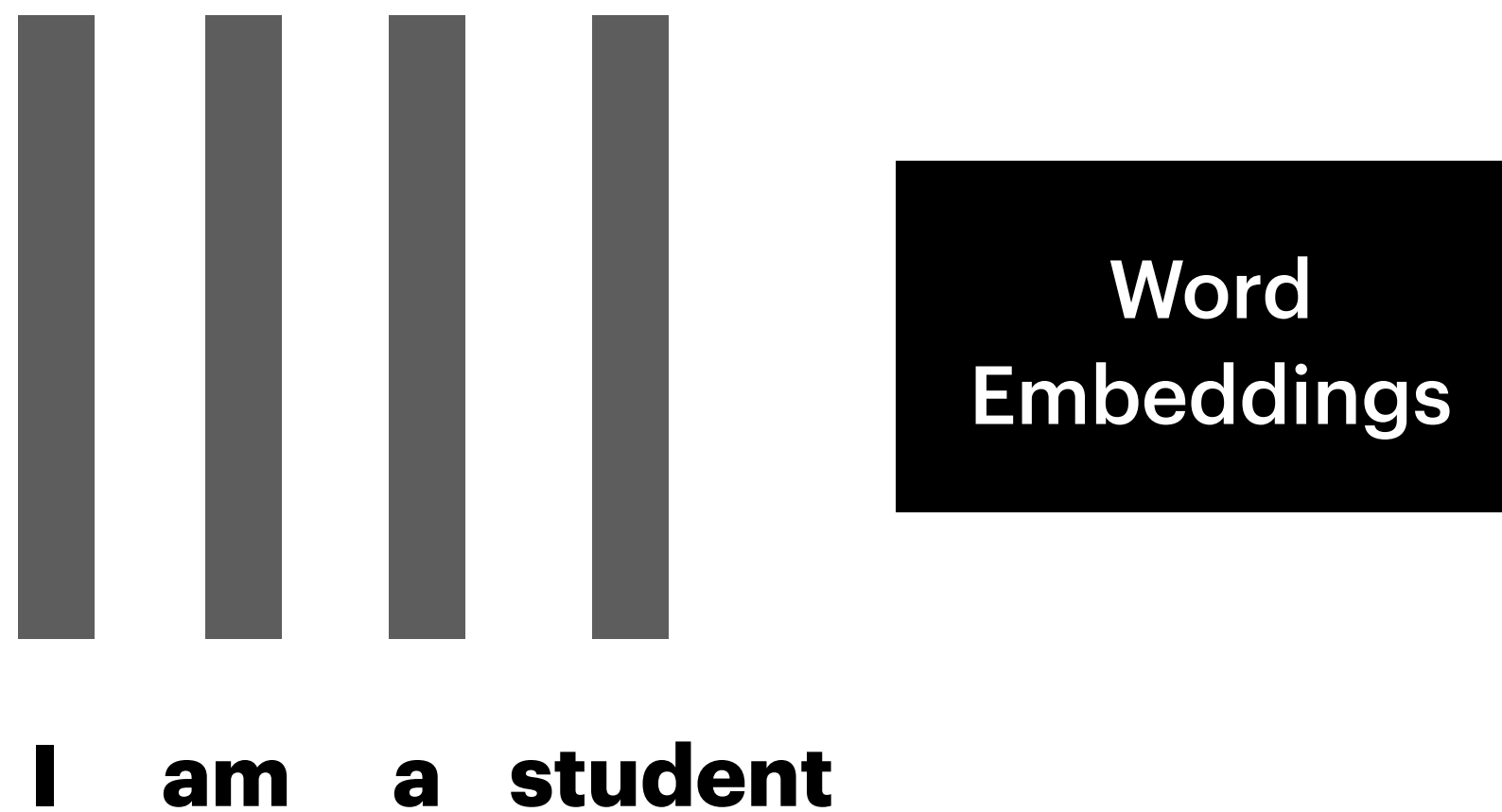
W_1

word2vec visualization



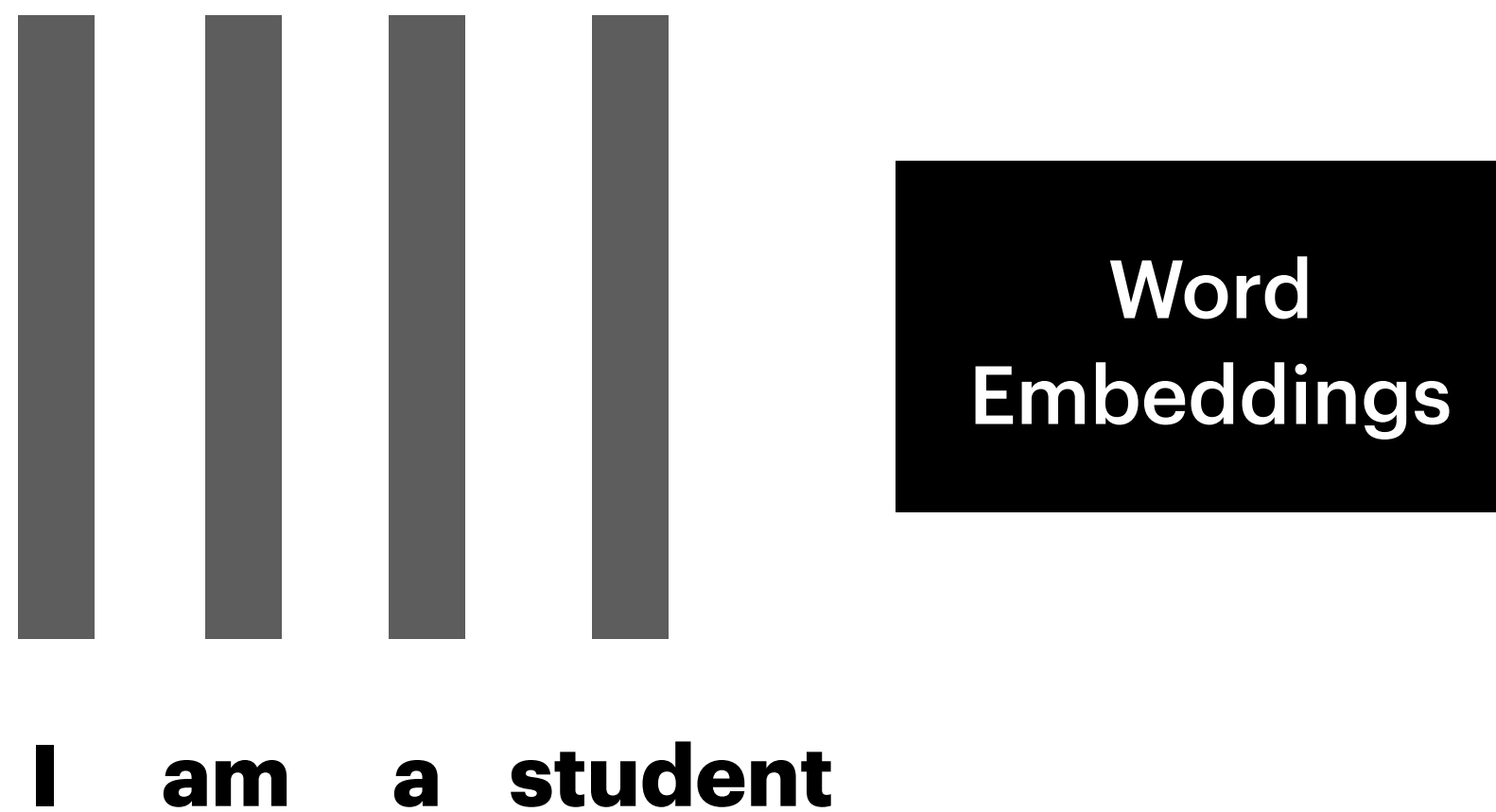
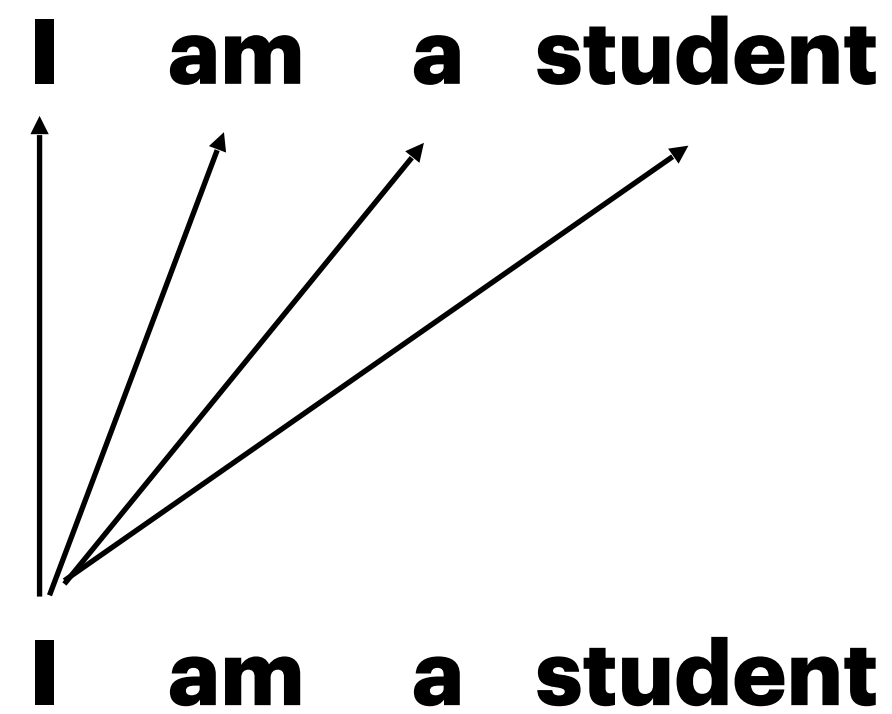
Transformers

- Embedding context in sequence inputs



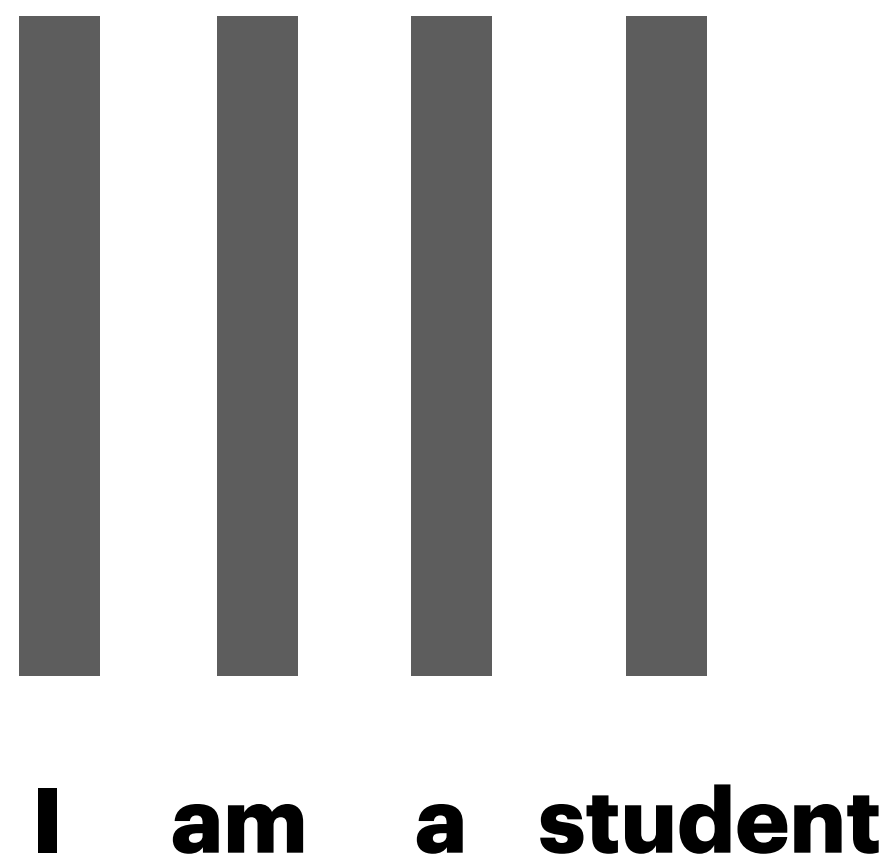
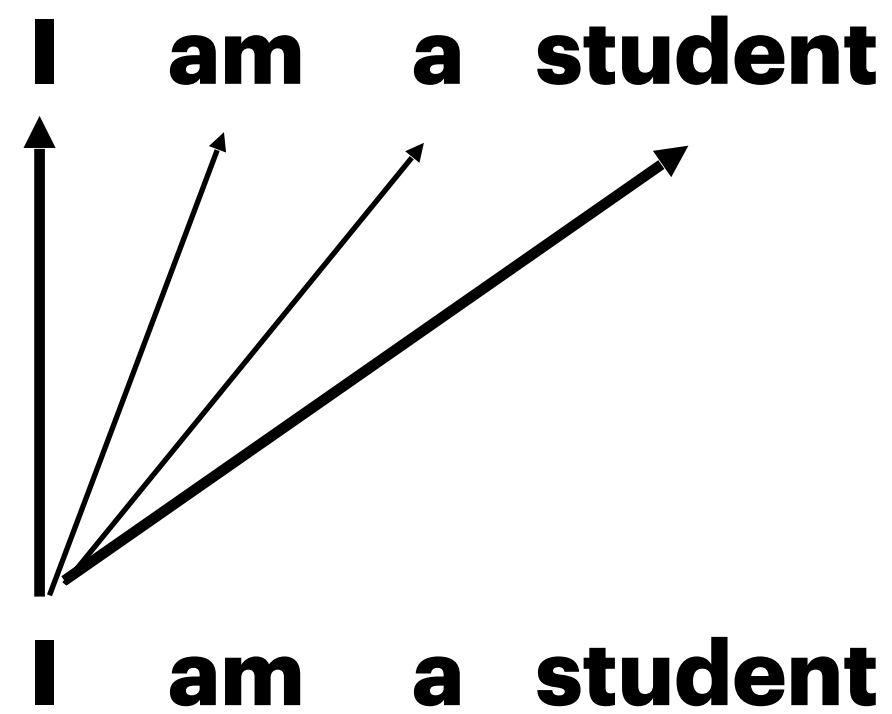
Transformers - self-attention

- Embedding context in sequence inputs
 - * Let us take an example



Transformers - self-attention

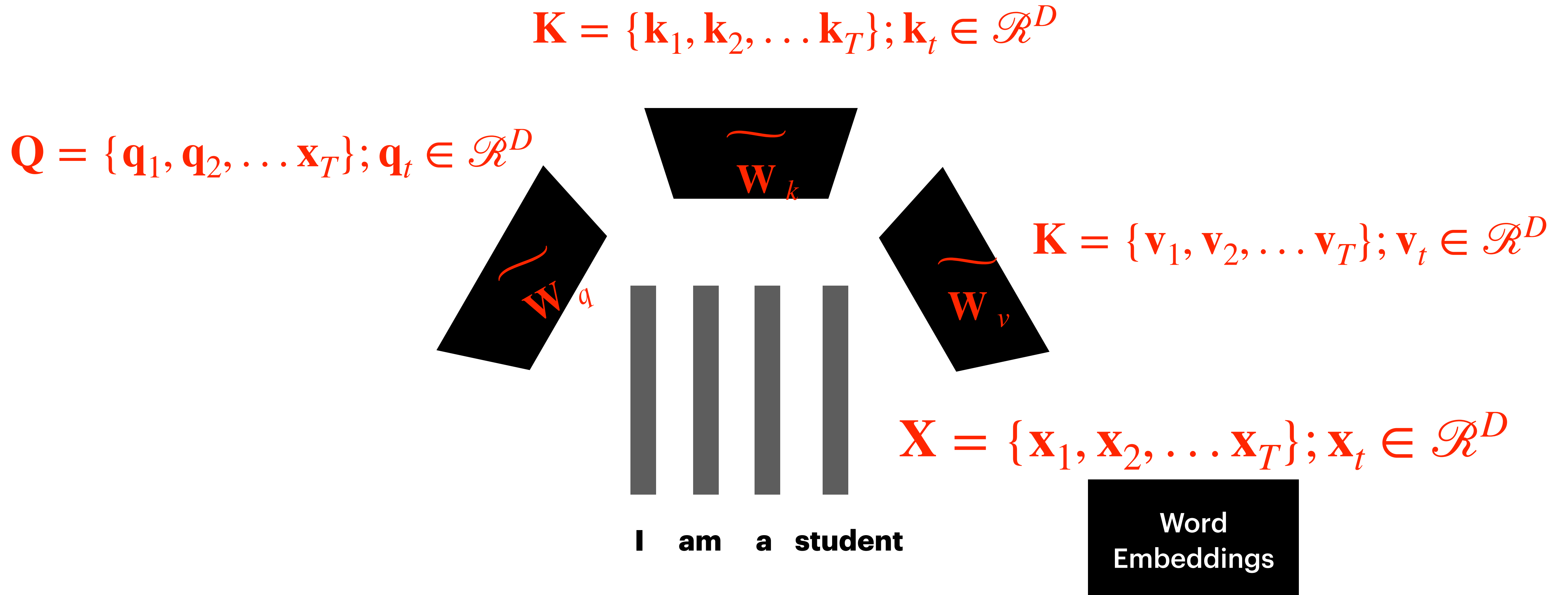
- Embedding context in sequence inputs
 - * Let us take an example
 - * Using word embeddings as the input representation



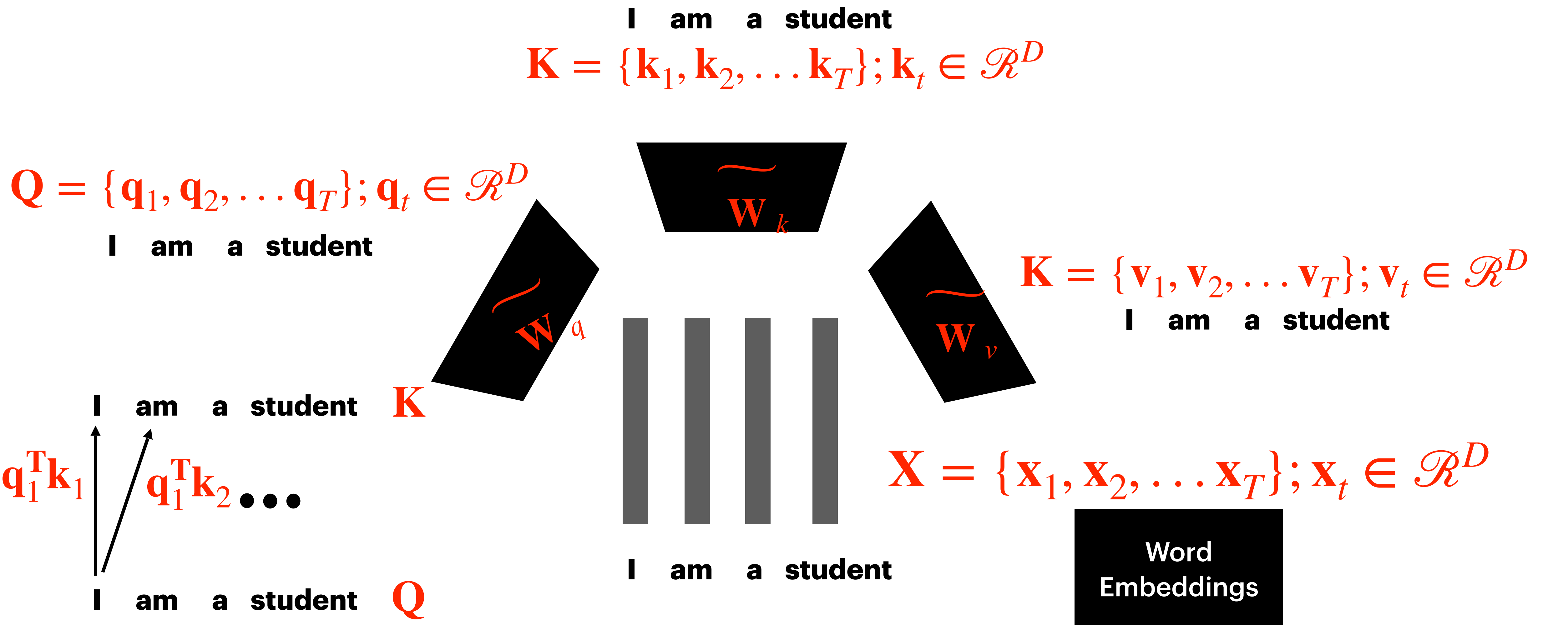
$$X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}; \mathbf{x}_t \in \mathcal{R}^D$$

Word
Embeddings

Transformers - self-attention



Transformers - self-attention



Transformers - self-attention

I am a student

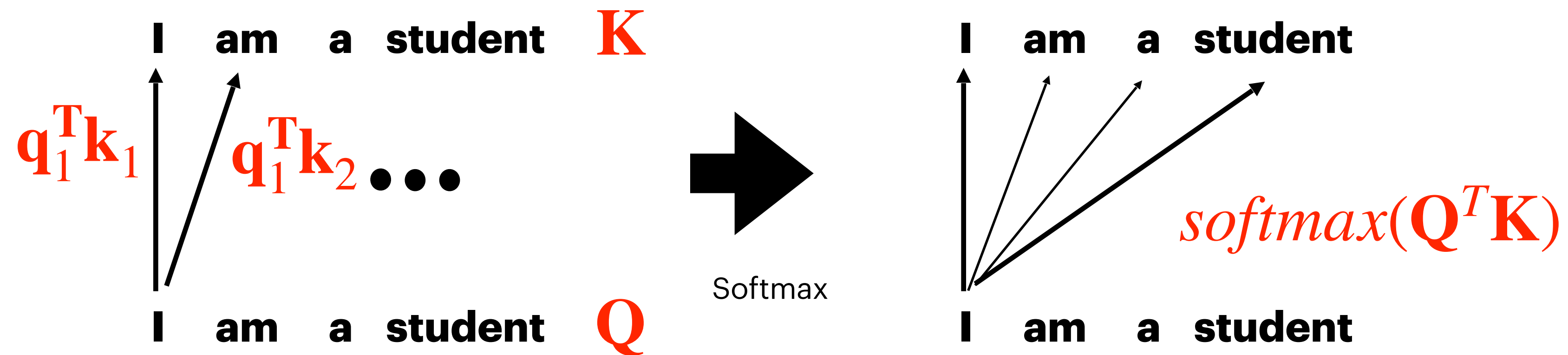
$$\mathbf{K} = \{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_T\}; \mathbf{k}_t \in \mathcal{R}^D$$

I am a student

$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_T\}; \mathbf{q}_t \in \mathcal{R}^D$$

$$\mathbf{K} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_T\}; \mathbf{v}_t \in \mathcal{R}^D$$

I am a student



Transformers - self-attention

I am a student

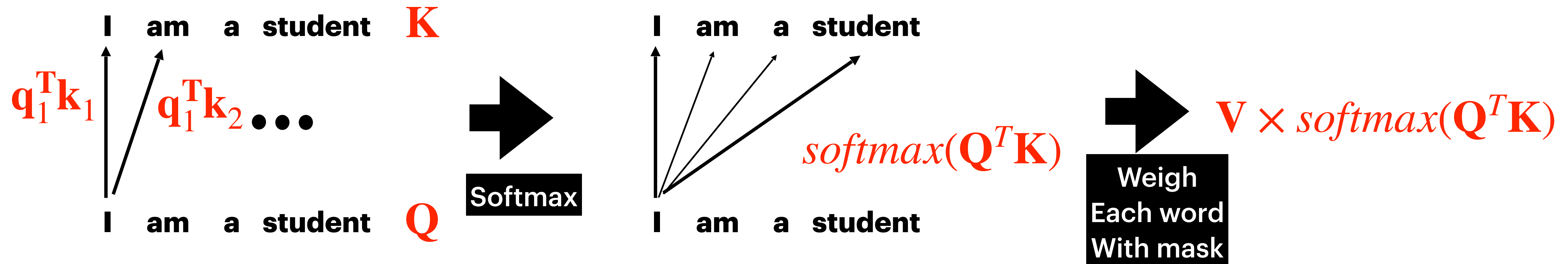
$$\mathbf{K} = \{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_T\}; \mathbf{k}_t \in \mathcal{R}^D$$

I am a student

$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_T\}; \mathbf{q}_t \in \mathcal{R}^D$$

$$\mathbf{K} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_T\}; \mathbf{v}_t \in \mathcal{R}^D$$

I am a student



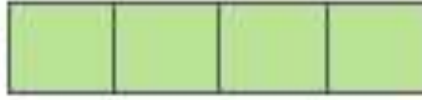
Transformers - self-attention

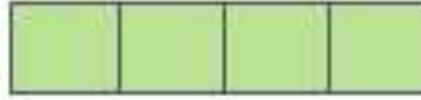
Input

Thinking

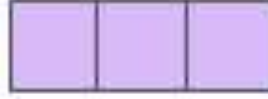
Machines

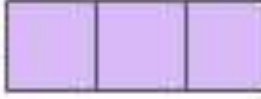
Embedding

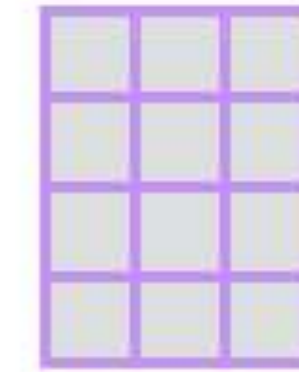
x_1 

x_2 

Queries

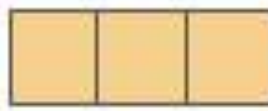
q_1 

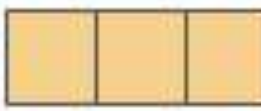
q_2 

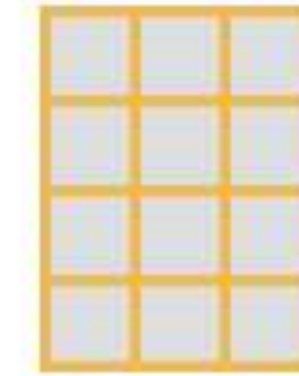


W^Q

Keys

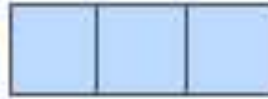
k_1 

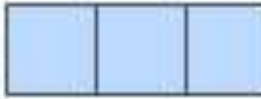
k_2 

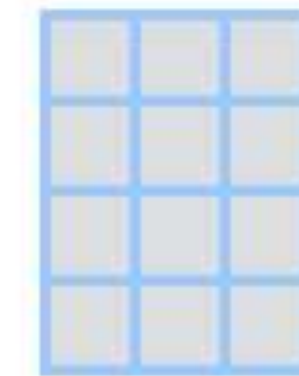


W^K

Values

v_1 

v_2 



W^V

Transformers - self-attention

Input

Embedding

Queries

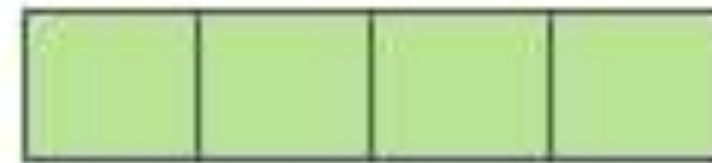
Keys

Values

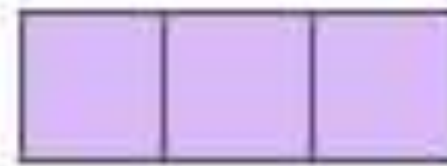
Score

Thinking

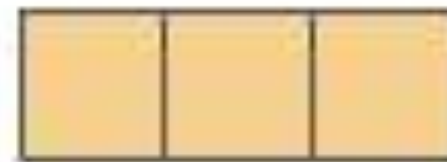
x_1



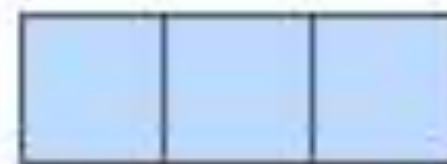
q_1



k_1



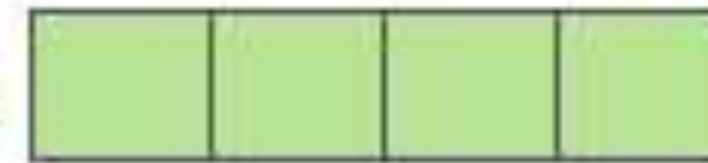
v_1



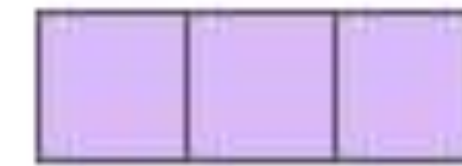
$$q_1 \cdot k_1 = 112$$

Machines

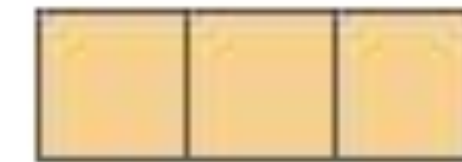
x_2



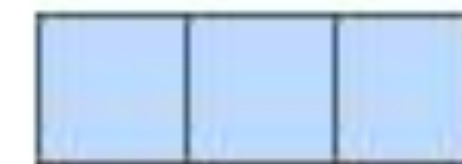
q_2



k_2

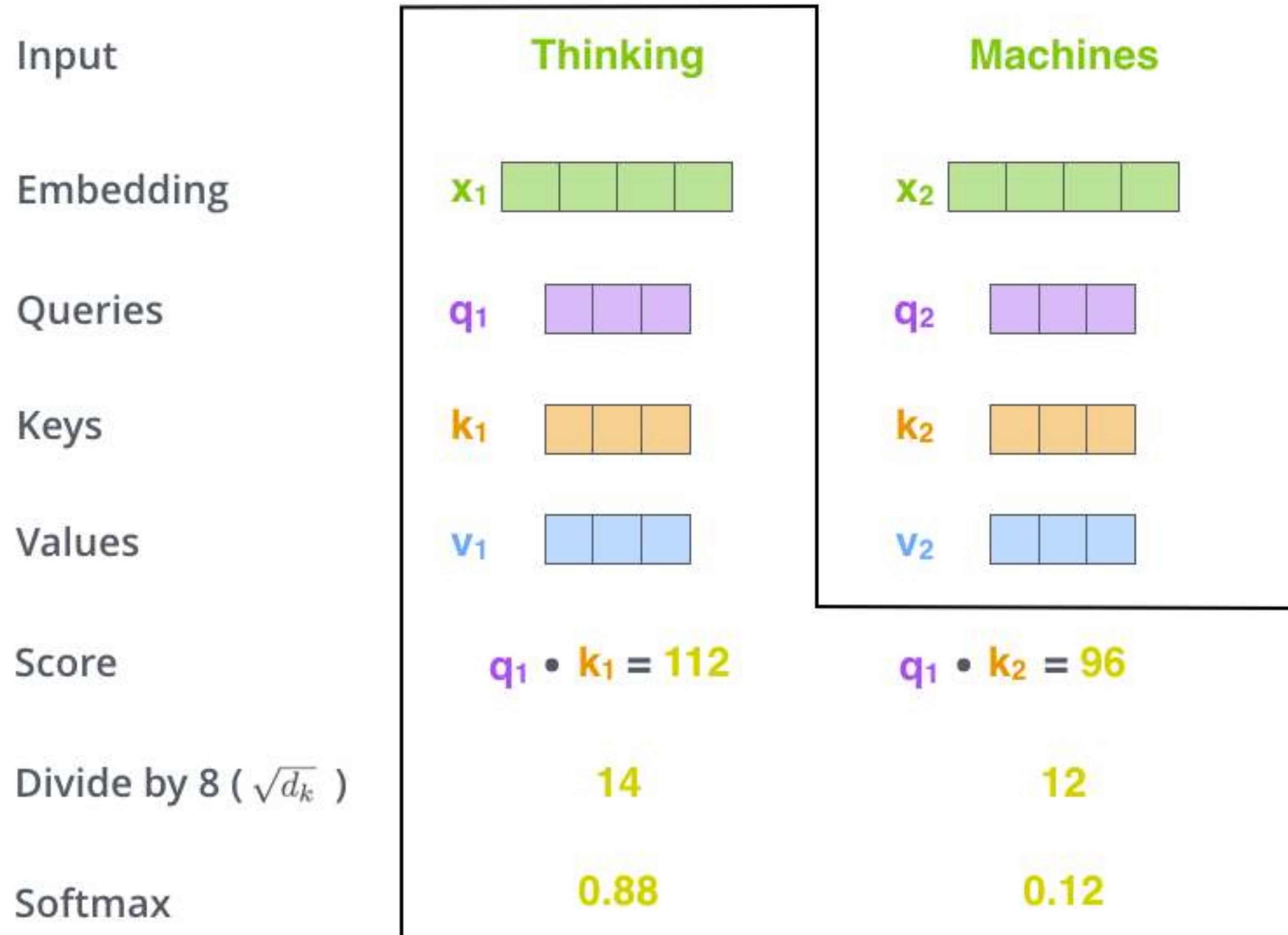


v_2

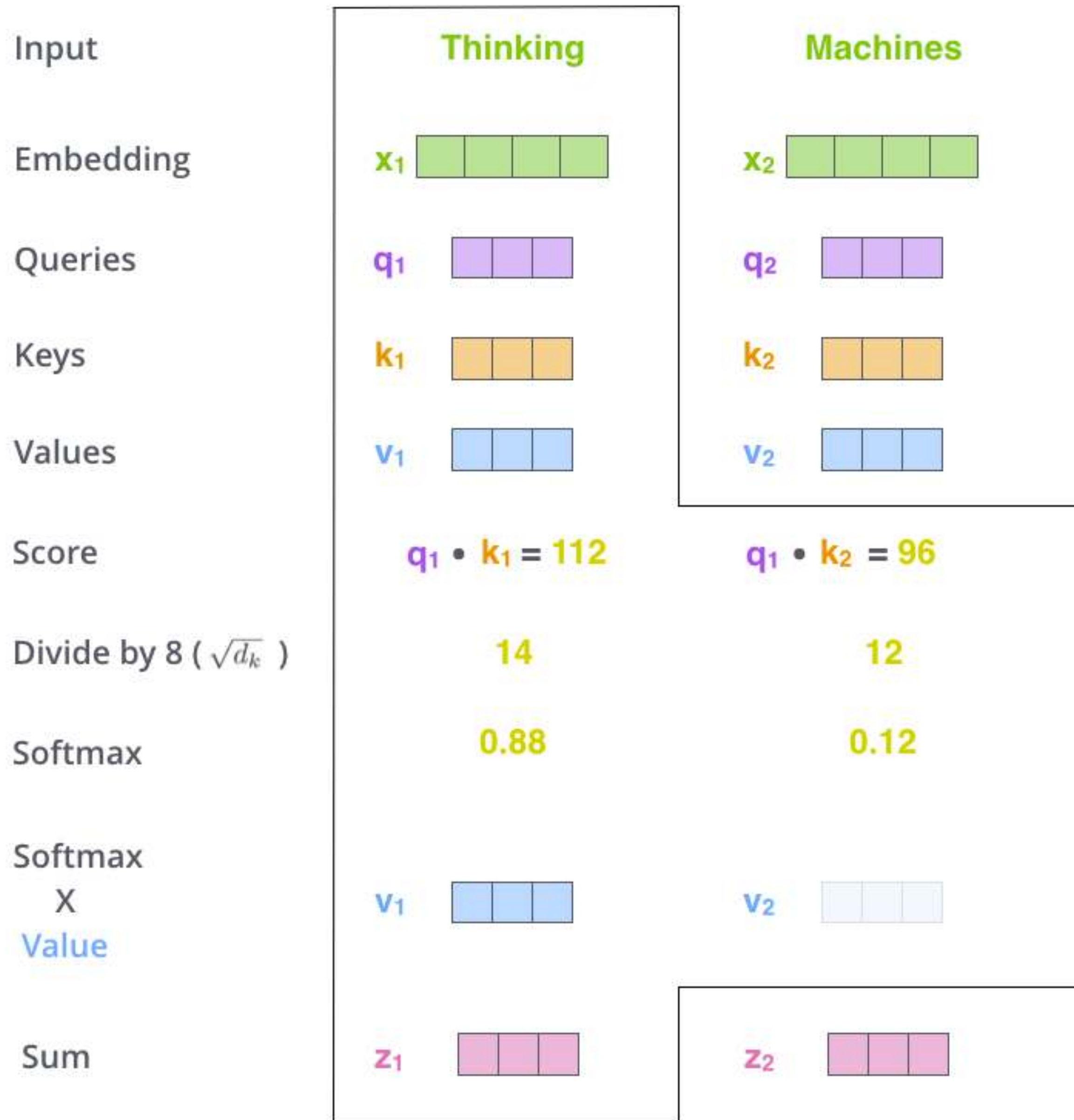


$$q_1 \cdot k_2 = 96$$

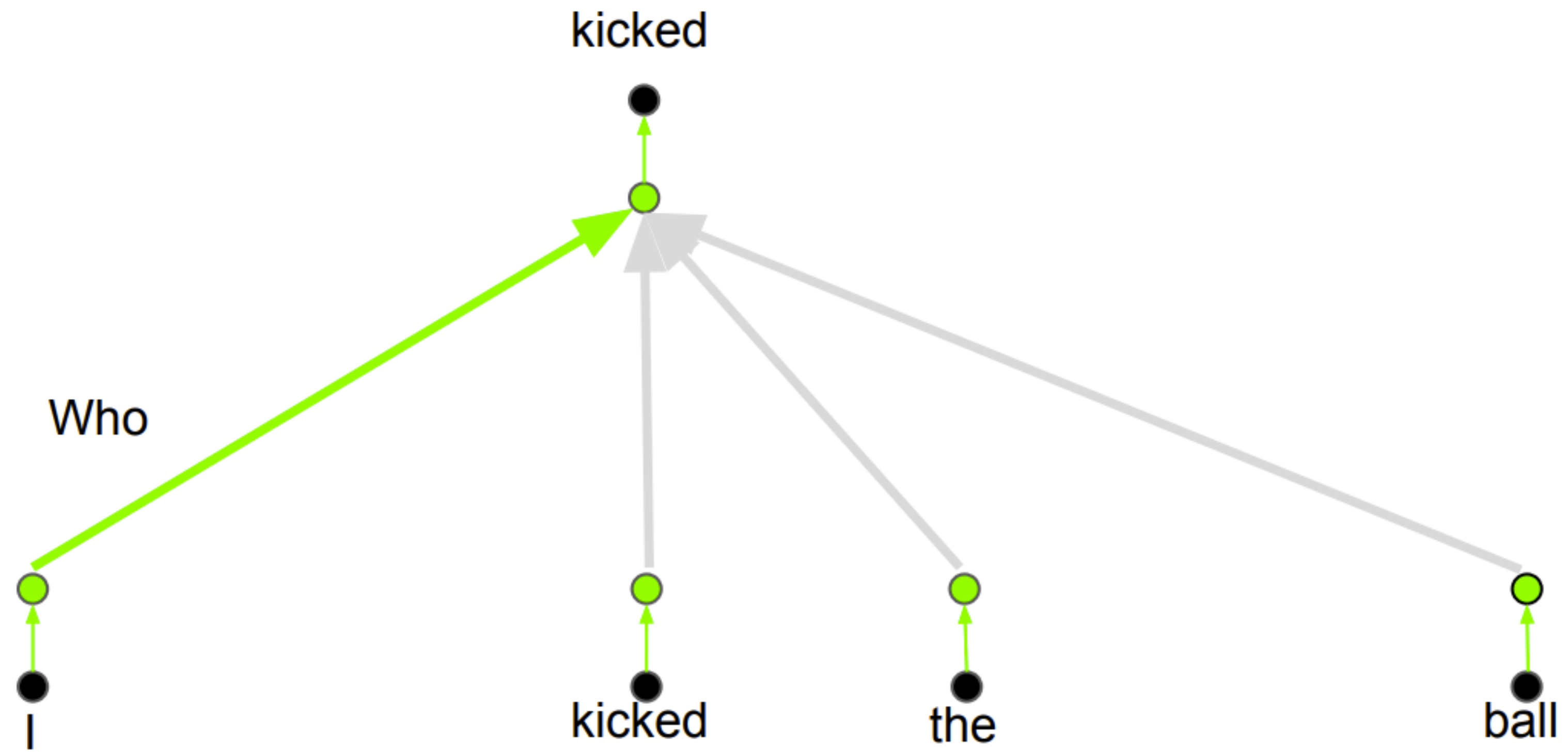
Transformers - self-attention



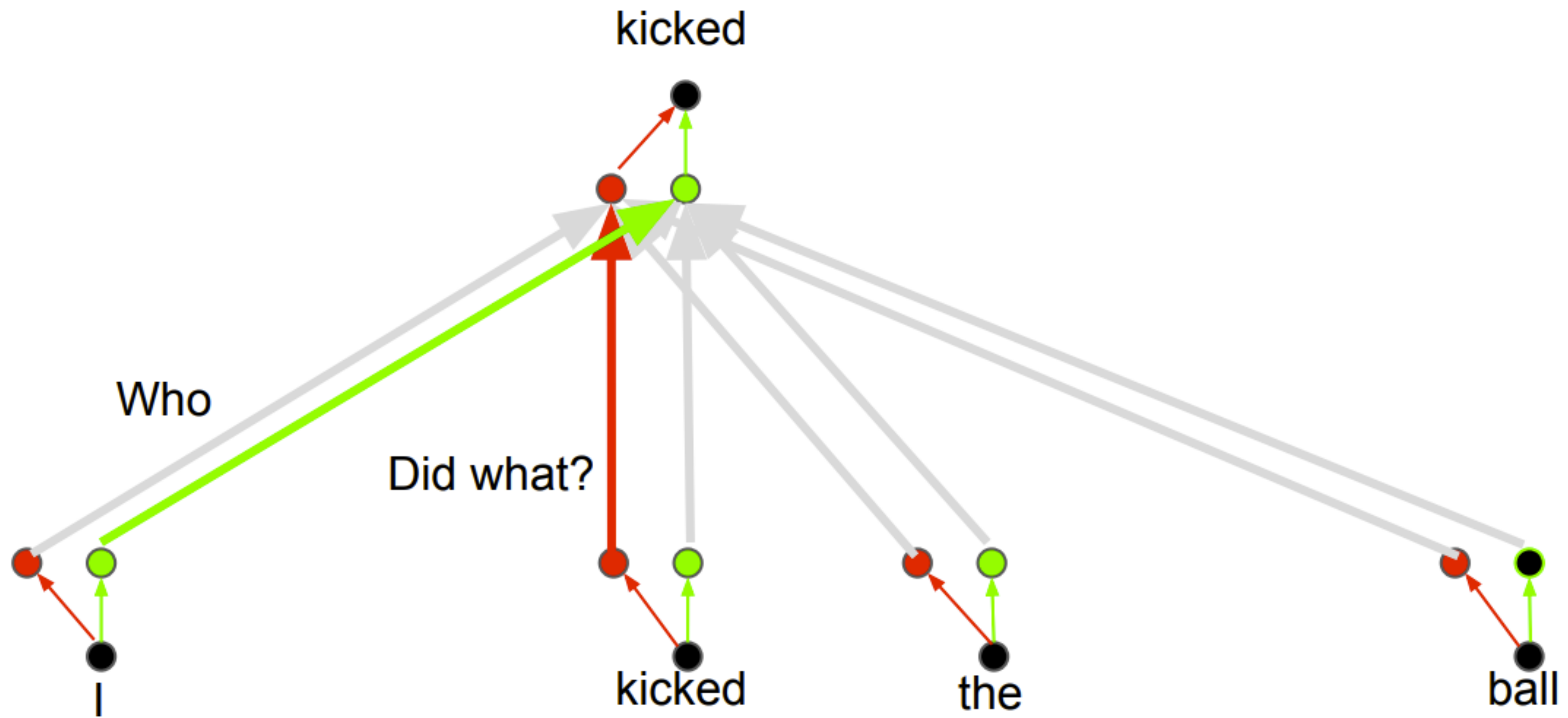
Transformers - self-attention



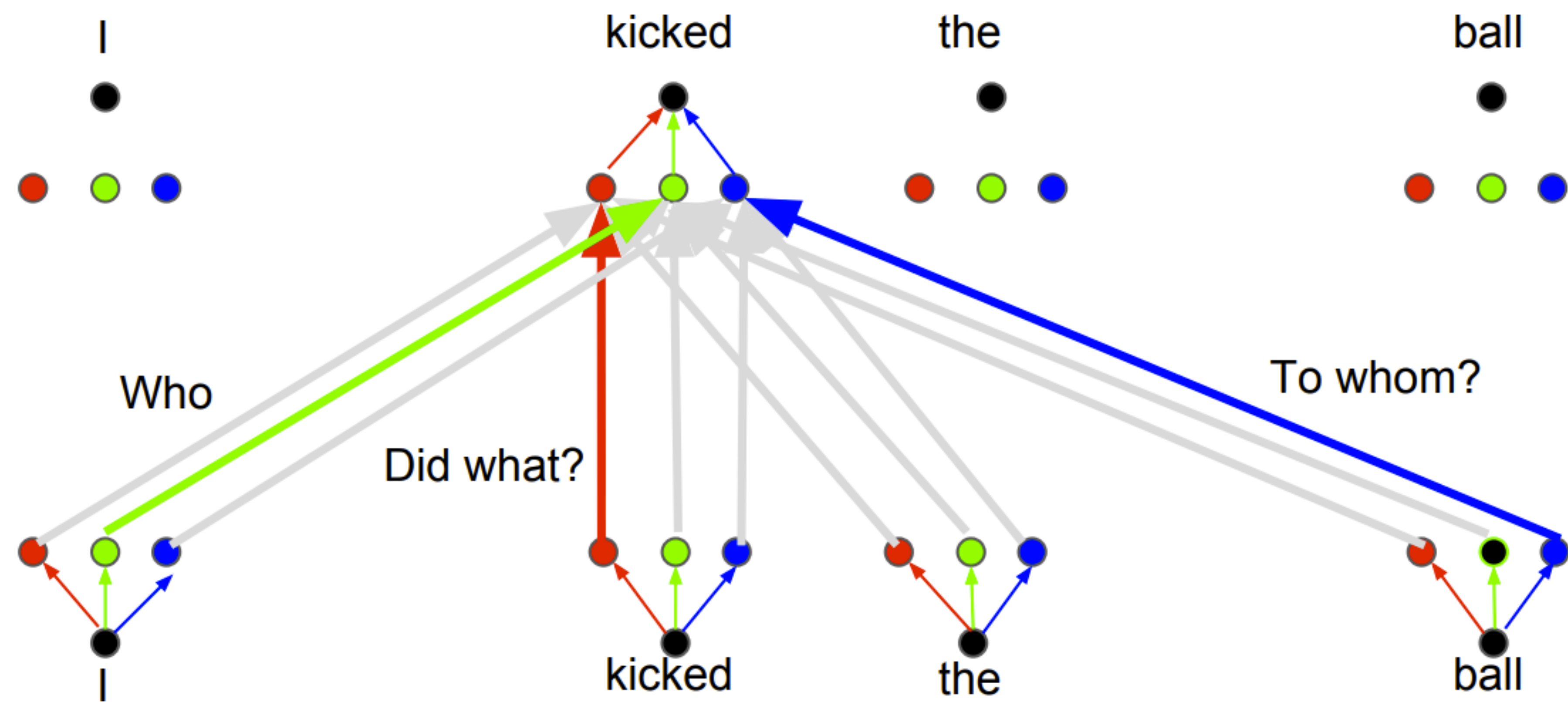
Multi-head



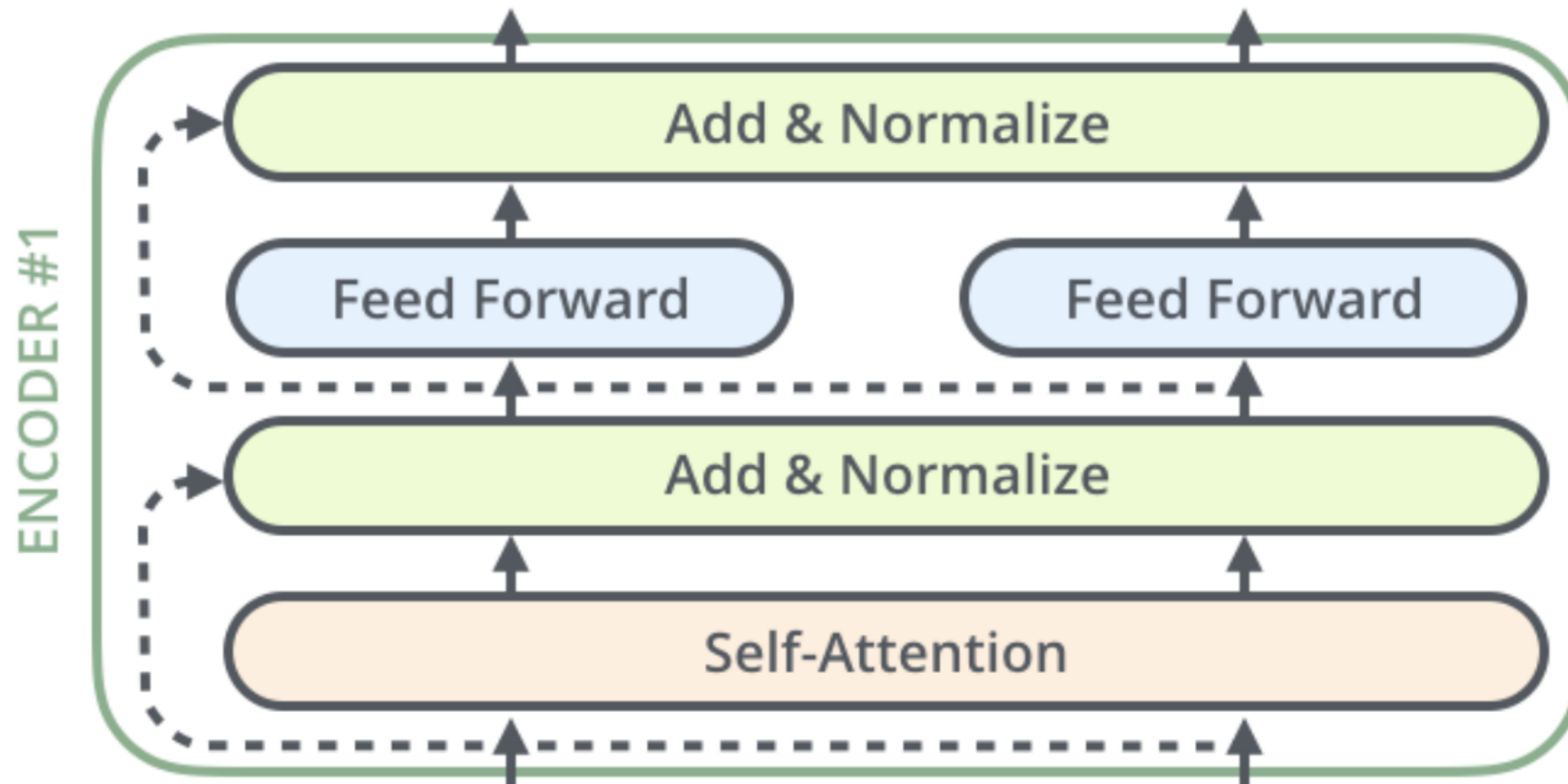
Multi-head



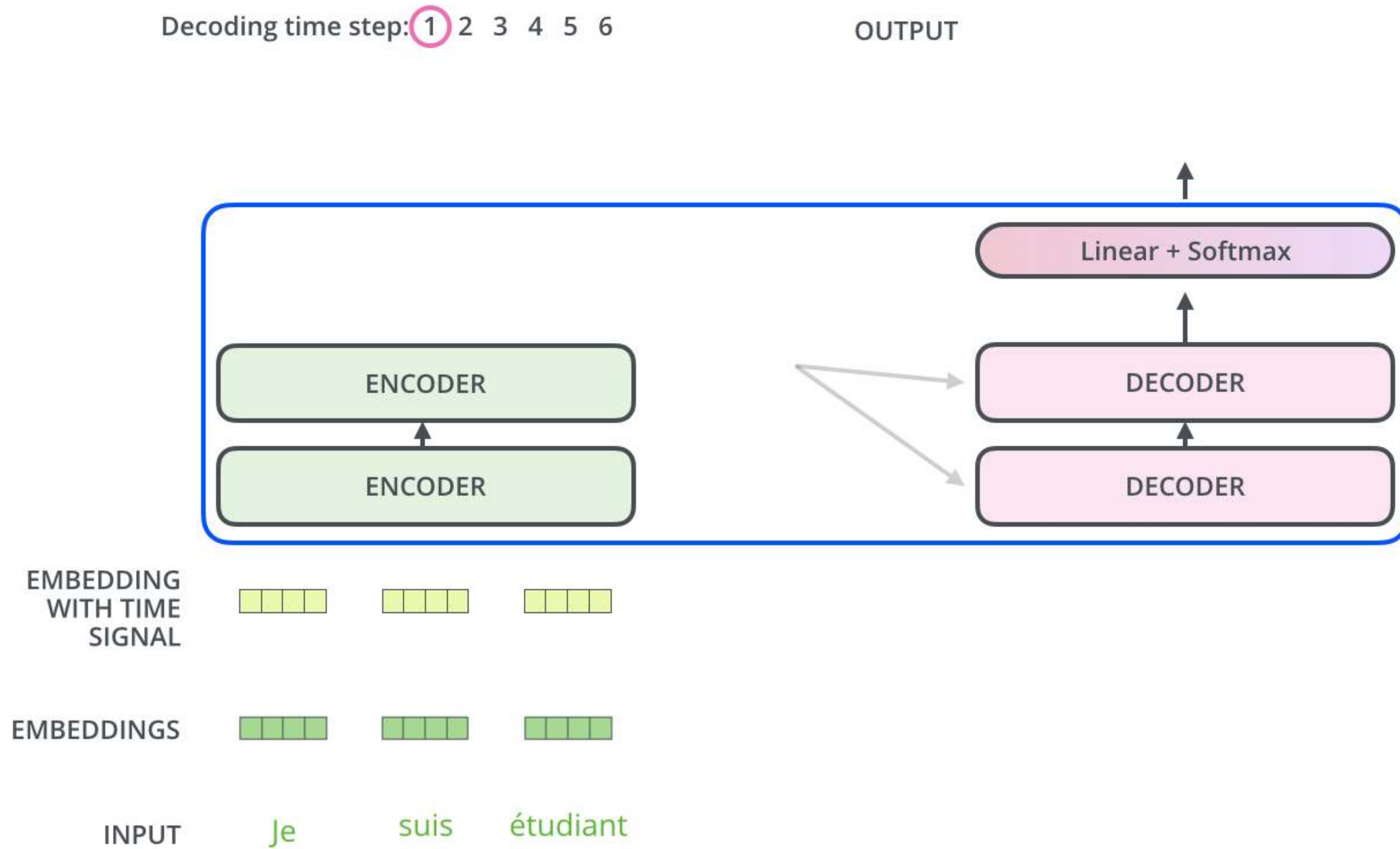
Multi-head



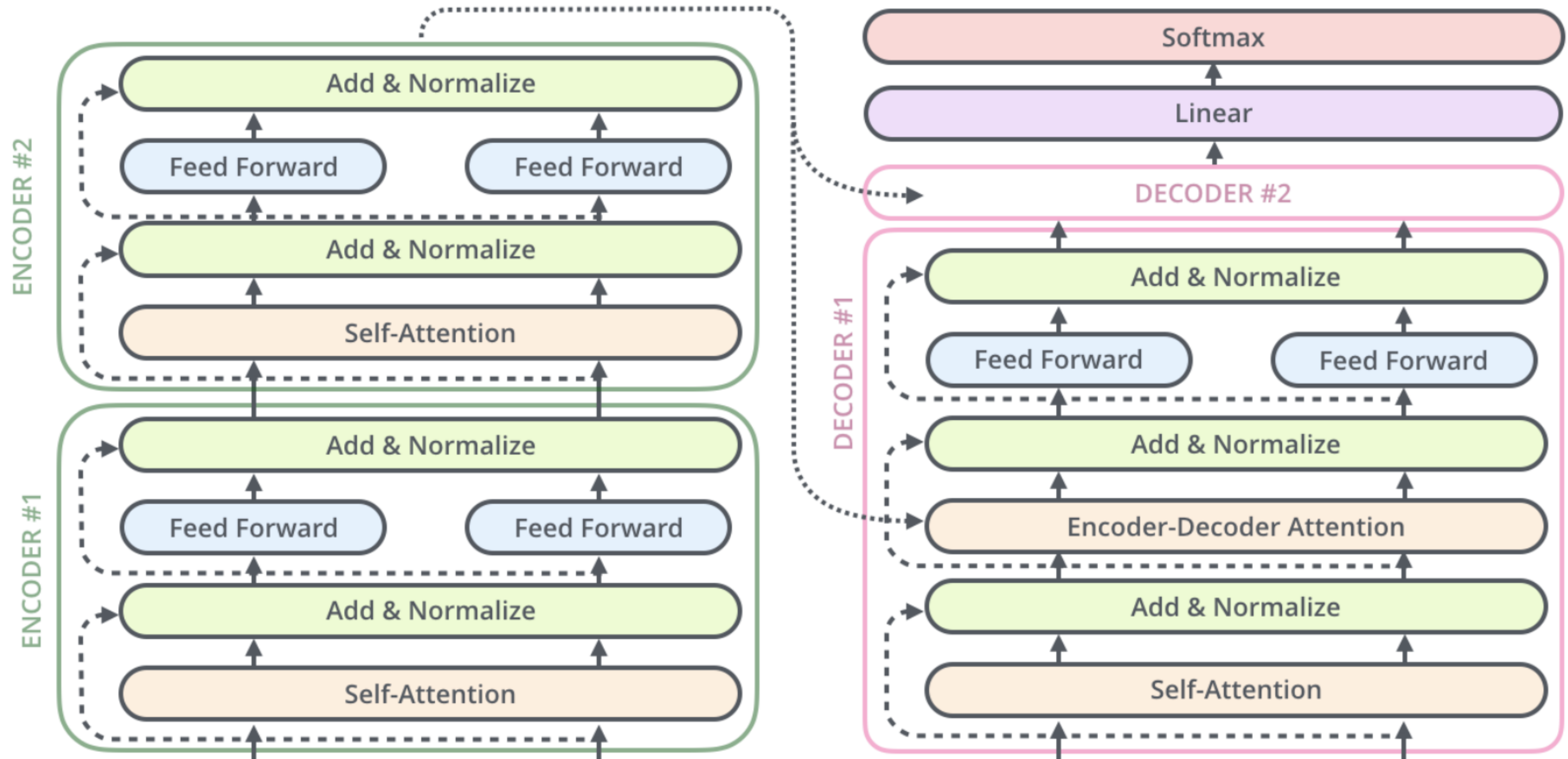
Transformer encoder



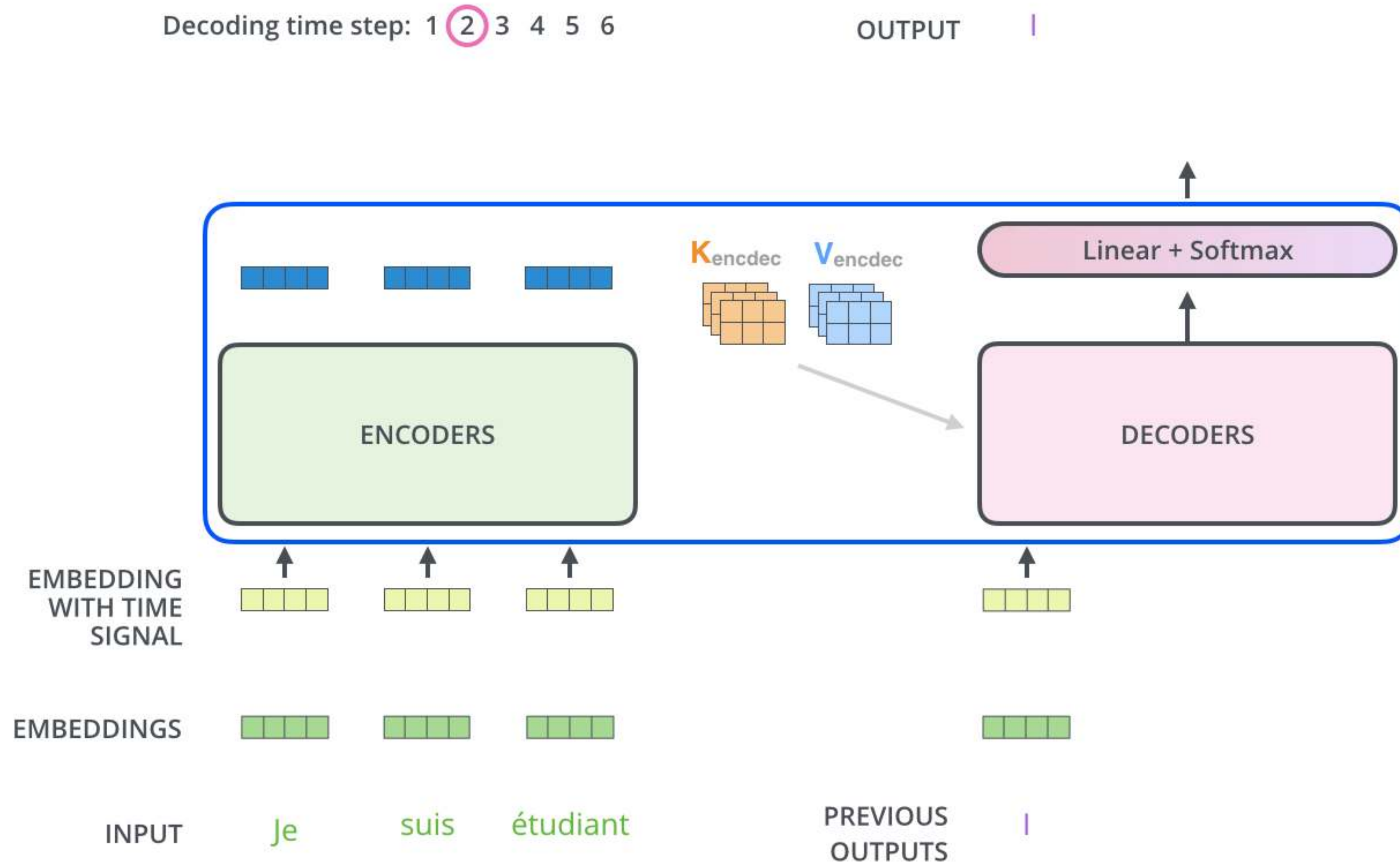
Transformer



Transformer decoder



Transformer Example

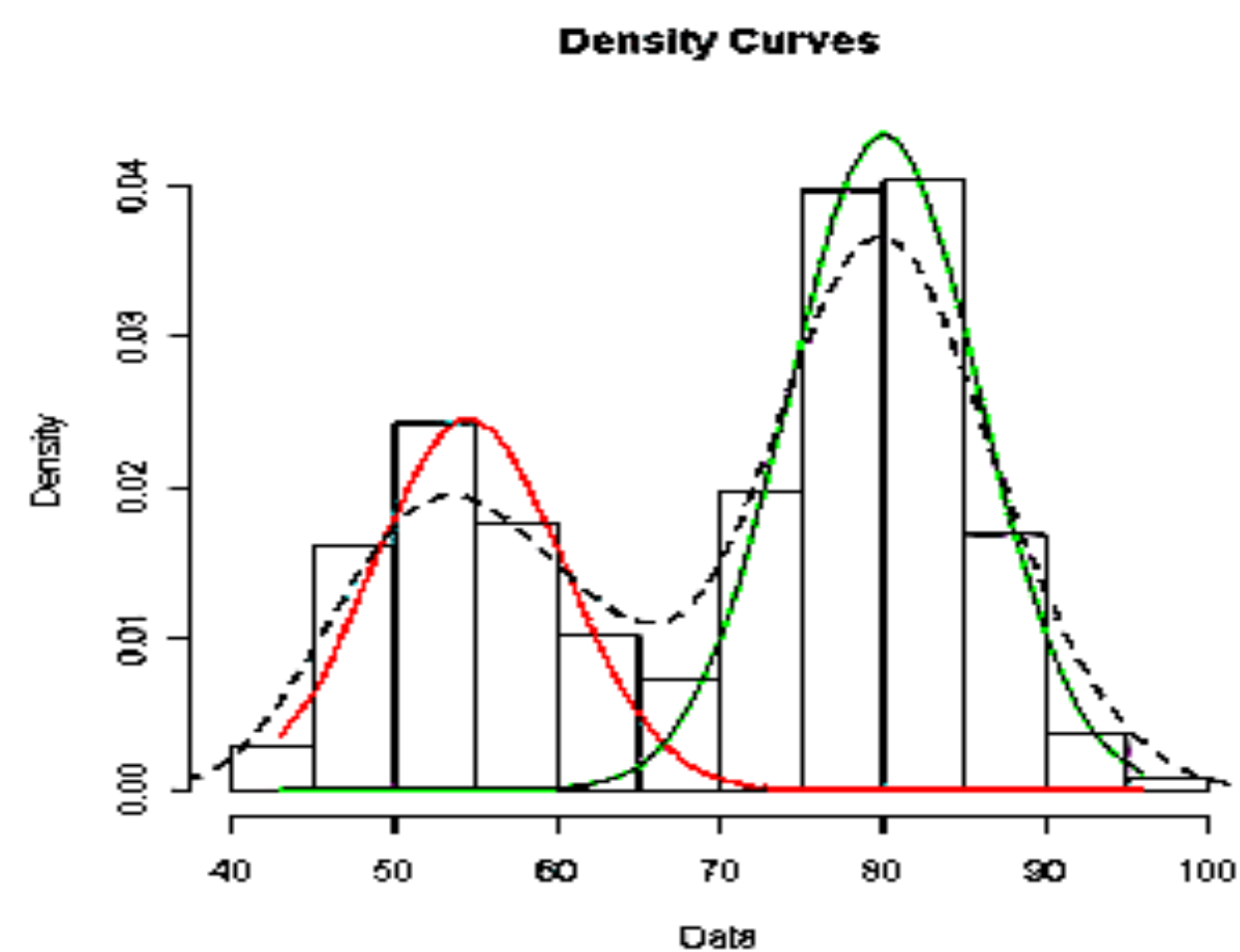


Unsupervised Learning

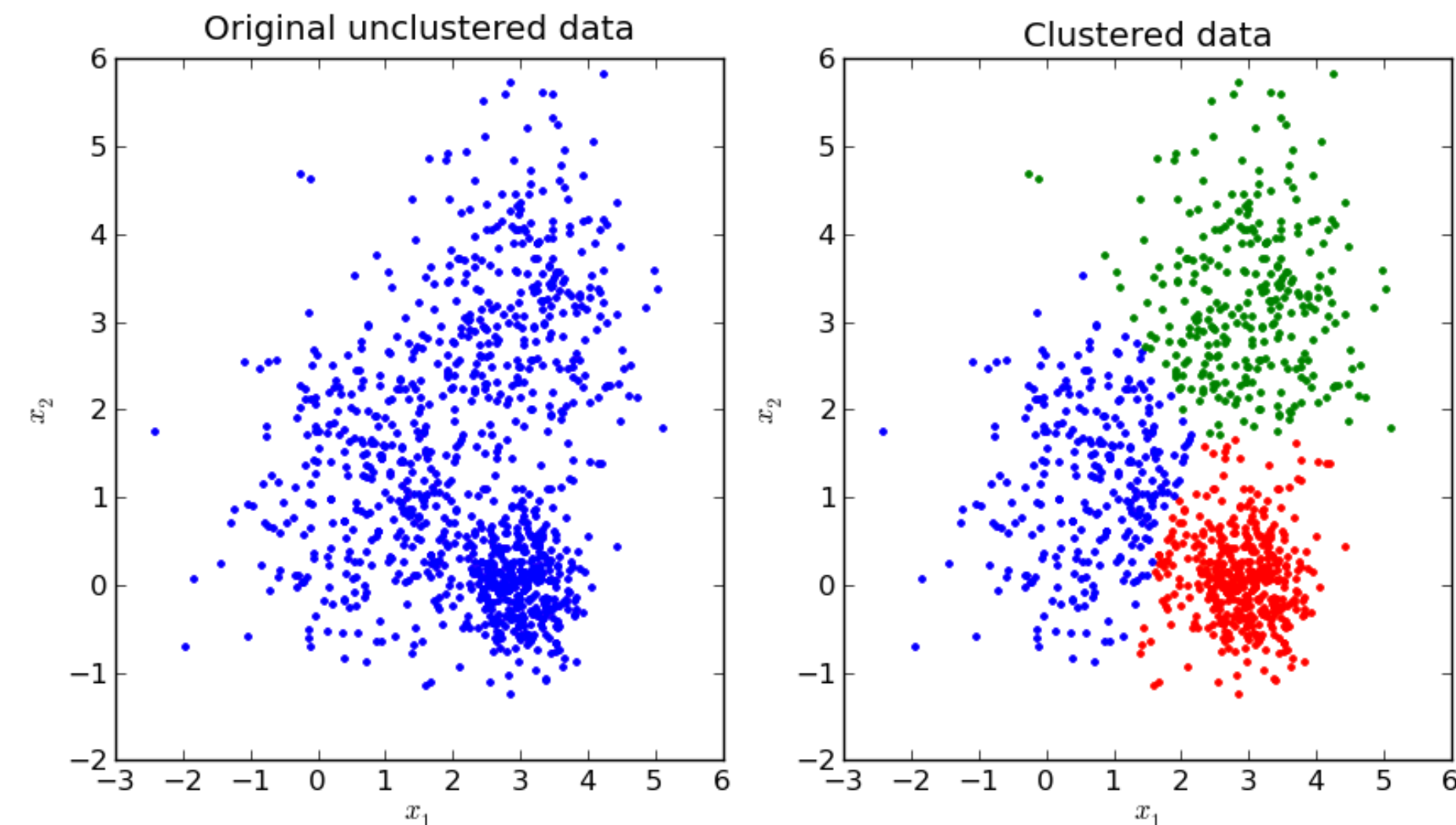
Unsupervised Learning

- Developing models that do not need labels
 - May model the generation of data.
 - May allow generation of new data samples
- Broad strategies for unsupervised learning

Learning the distribution of the data

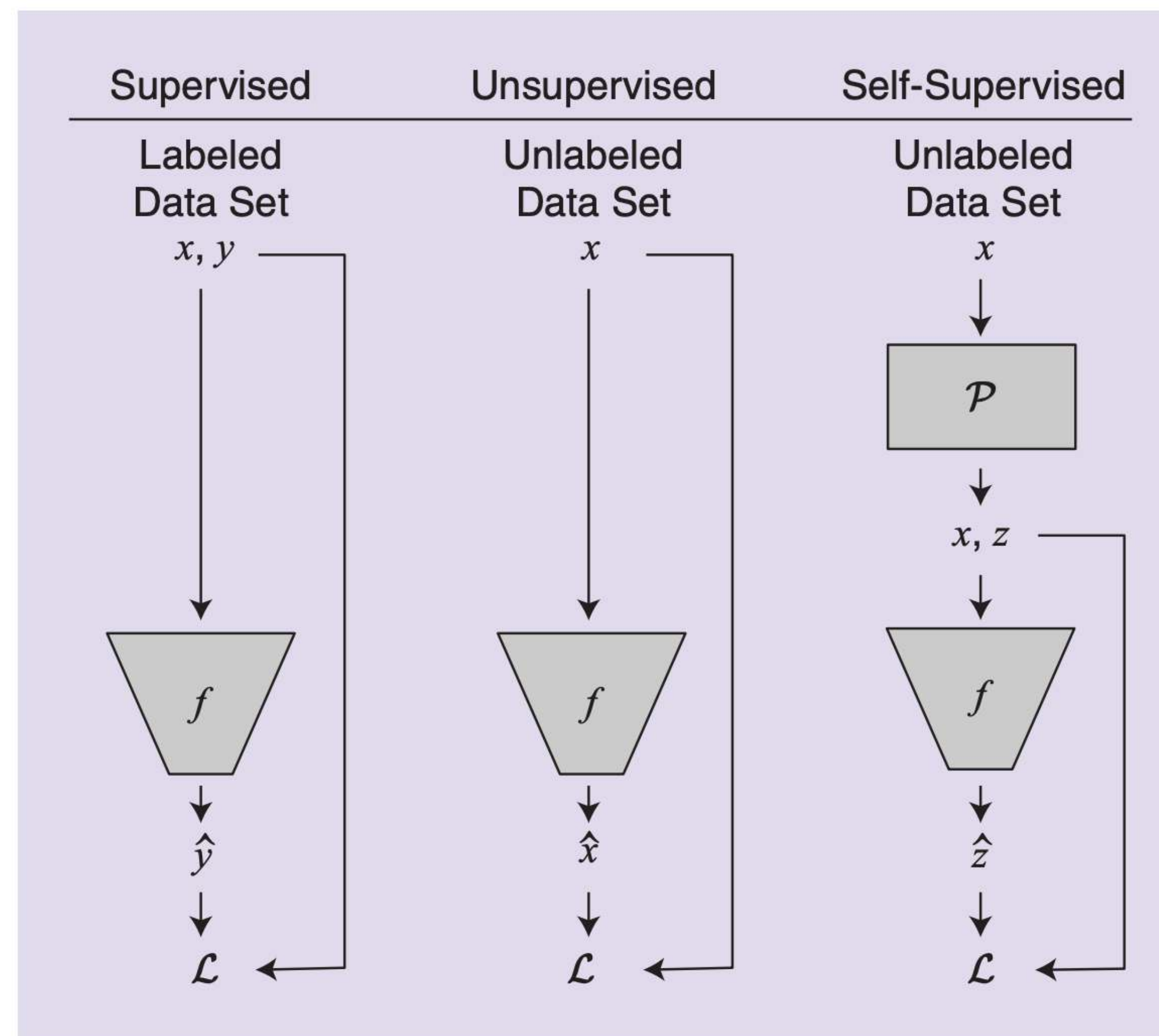


Detecting clusters in the data



Self supervision

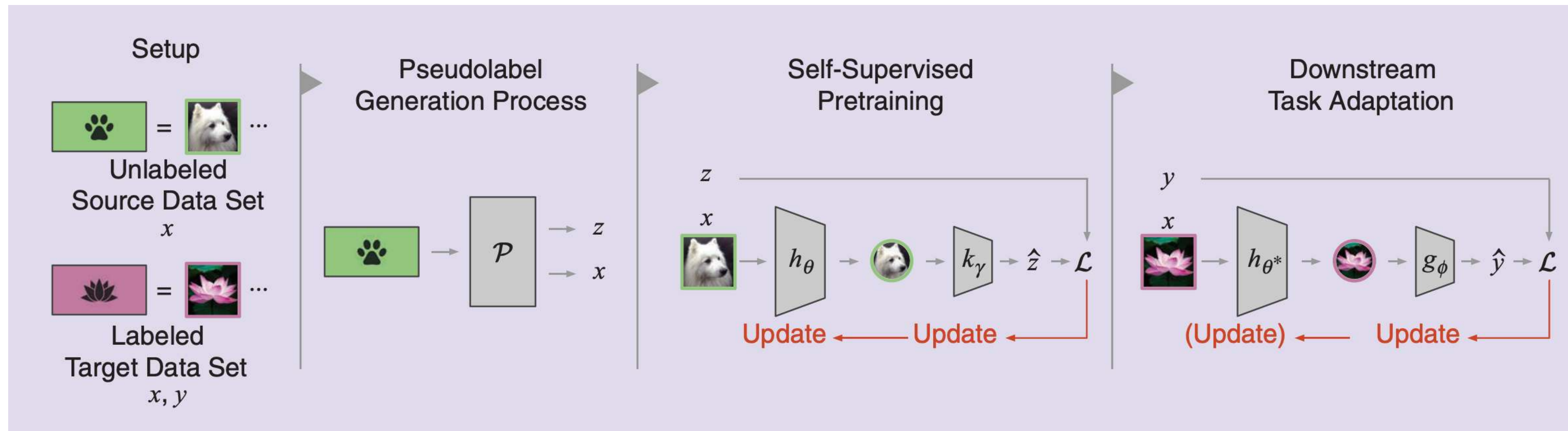
- ◆ Different from supervised and unsupervised learning
- * Does not perform distribution learning or reconstruction
- * Uses a pretext task
- * Performing contrastive or predictive learning
- ◆ Using large volumes of unsupervised data



Ericsson, Linus, et al. "Self-supervised representation learning: Introduction, advances, and challenges." *IEEE Signal Processing Magazine* 39.3 (2022): 42-62.

Self supervision - principle

- ◆ Two levels of modeling with unsupervised data
 - ❖ Generating a pseudo-label
 - ❖ Learning the upstream model
- ◆ Downstream task performs fine-tuning of the SSL model.

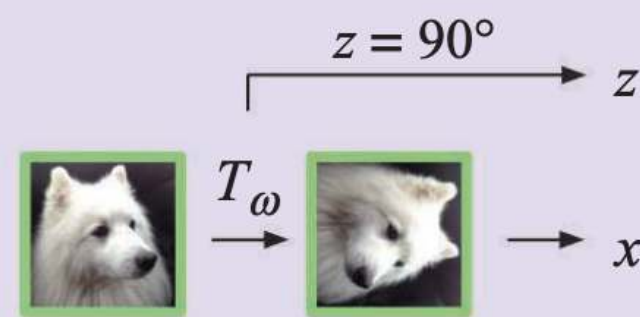


Self supervision - pre-text task

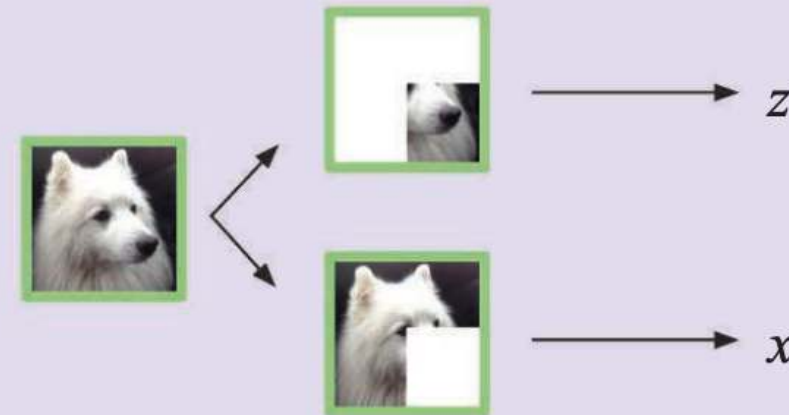
Pseudolabel Generation Processes



Transformation Prediction



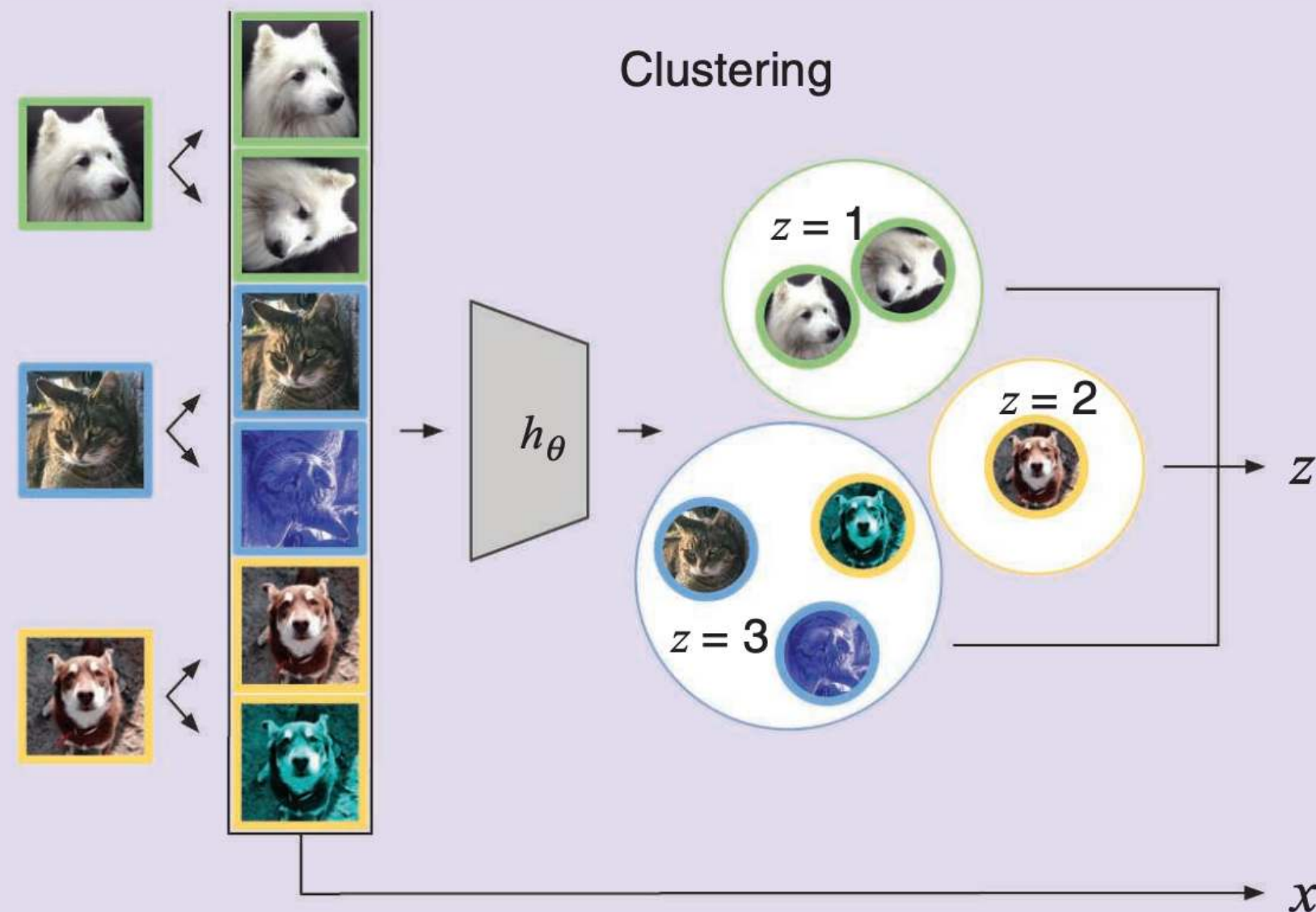
Masked Prediction



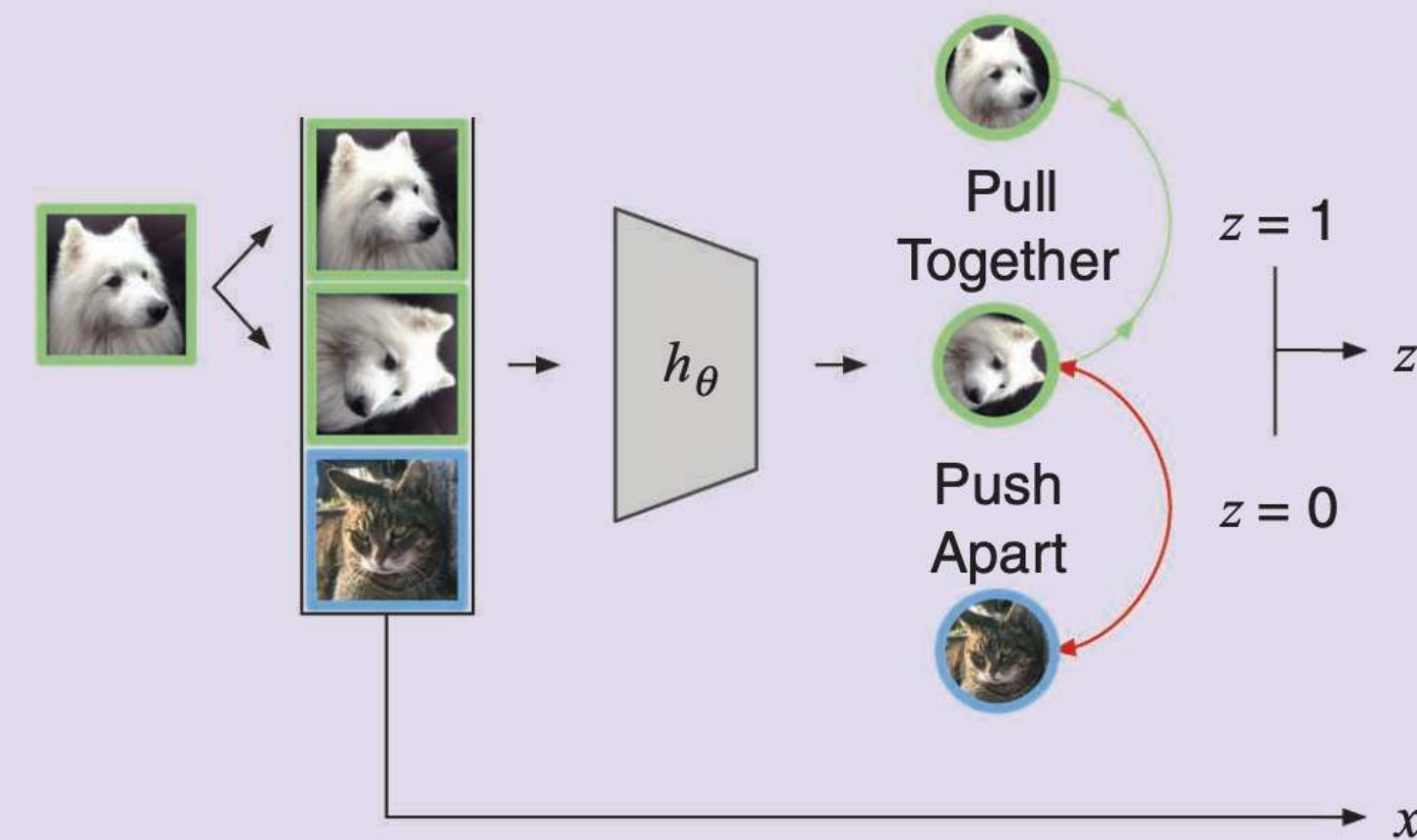
Instance Discrimination



Clustering

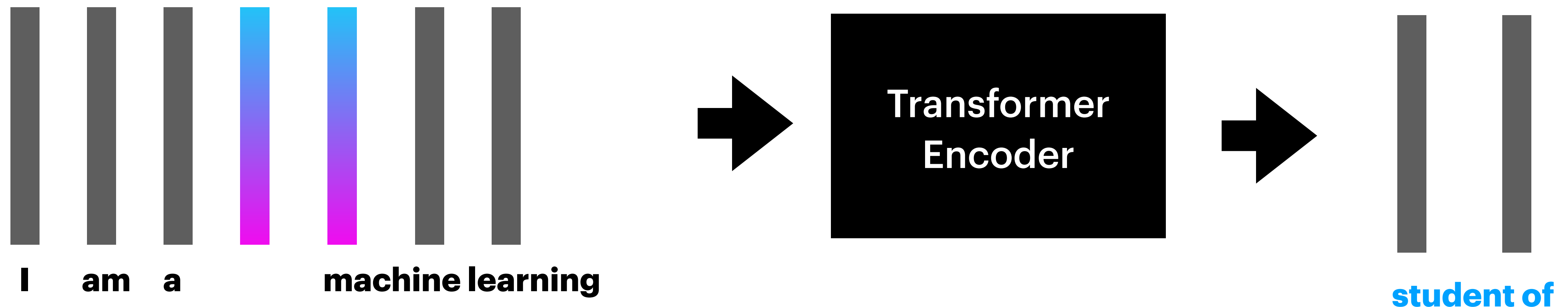


Contrastive Instance Discrimination



Self-supervision as a task

- Masking out portions of the input data
 - * Pass the rest of the embeddings (with zeros or random entries at the masked locations) to the transformer encoder
 - * Have the model predict the word tokens in the masked portions - **Masked Language Modelling (MLM)**



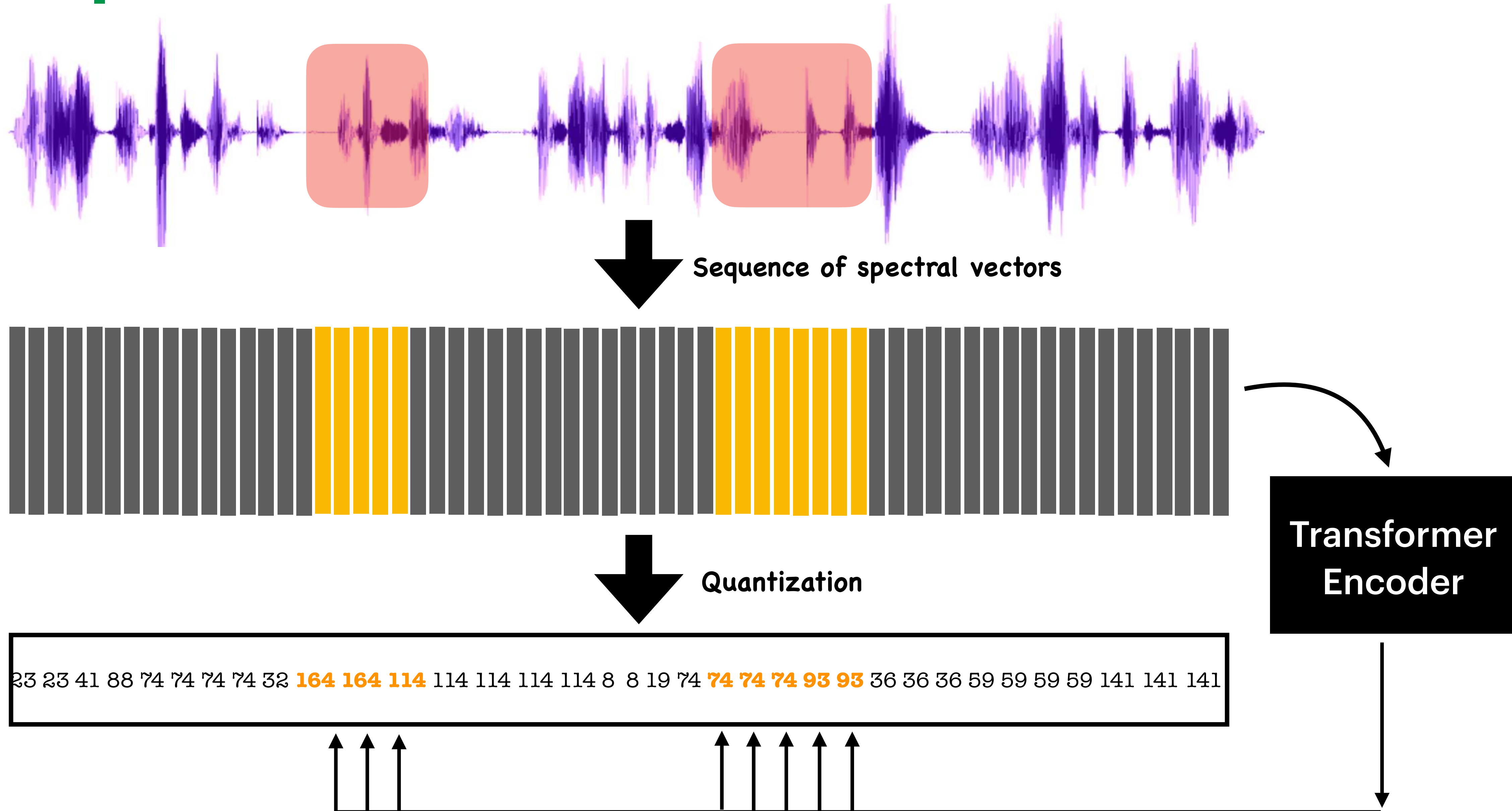
Large language models (LLMs)

- Extending the task of self-supervision
- Mine lots of text data
 - * Crawled from the web, as well as, from other resources.
- Design the **model with large capacity** (Millions → Billions of parameters)
- Pre-train the model
 - * With MLM and similar style of losses
 - * High resource of computations.
- Final trained model can be **fine-tuned for supervised tasks**
 - * Load the parameters as initialization and perform supervised learning.

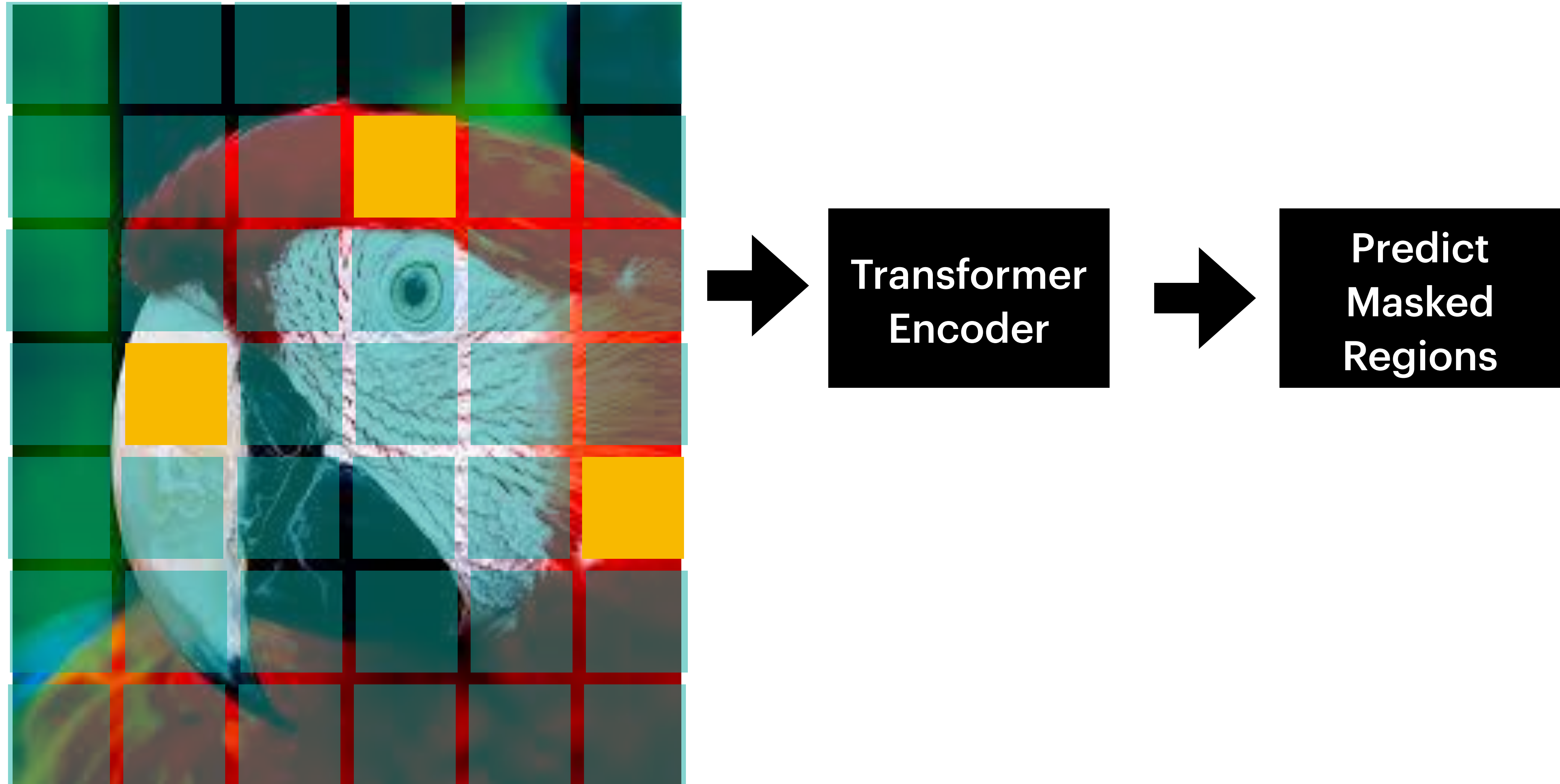
Large language models (LLMs)

- Self-supervised learning
 - * Has shown emergent abilities to generalise to wide variety of downstream tasks.
 - ✓ Tasks that the model was not trained on
 - ✓ Not seen in smaller models
 - * Enables to build reasoning capabilities in the model.
 - * **Applicable for several domains** – text, speech and images.

Self-supervision in audio - wav2vec



Self-supervision in images - Vision Transformer



Future works (some already underway)

- Multi-modal
 - * Incorporating learning across modalities
 - ✓ Create a domain specific encoder/decoder and learning the joint language model.
- Combining some labeled data with the self-supervised data to further improve the models.
 - ✓ Current models like GPT use human feedback.
- Understanding the risks and vulnerabilities of these models.

THANK YOU

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